

ANTHROPIC

Appendix to: “What work can robots do?”

September 30, 2026

A. Measuring exposure

Our analyses use data from O*NET, a database of around 900 occupations linked with descriptions of around 19,000 job tasks. We first use Claude to classify tasks as physical, cognitive, or interpersonal. Our analysis focuses on physical tasks, which we think require robots to automate. Though humanoid robots might perform cognitive and interpersonal work too, automating most of this work doesn't require better robots per se. We then assess whether robots today can perform each physical task and under what circumstances by querying Claude with web search. Full prompts appear in Appendix F.

A.1 Task taxonomy

We decompose tasks into physical, cognitive, and interpersonal work, following a long tradition in labor economics and job analysis. For example, the job taxonomy in the Dictionary of Occupational Titles categorizes work by whether it acts on “Data/People/Things.”

We define three variables to describe the physical, cognitive, and interpersonal components of job tasks. For component $k \in \{physical, cognitive, interpersonal\}$ and task t , define

- $time_{tk}$: time for task t spent on k work. Times sum to 100.
- $level_{tk}$: level of skill on component k required to perform task t , 0-5. This scale measures the practice a person needs to reach that level, from none (1) to elite performance that few trained adults reach (5). Level 0 means the task makes only incidental use of that component.
- $essential_{tk}$: indicator for whether task t could not be completed were component k removed.

We use Claude to rate each O*NET 29.3 task on these dimensions, providing it with the task description and the title and description of the occupation that performs it. Since O*NET tasks can be quite broad, we prompt Claude to describe specific instances of tasks that a typical US worker would encounter. For example, consider the nondescript task “Perform teller duties as required” for new accounts clerks. Claude notes that this work involves handling a “suspected counterfeit bill, ... or a frustrated customer disputing a fee,” as well as “counting currency by hand ... buying and selling cash to the vault ... pulling

the pneumatic tube canister or deposit bag.” In turn, this task is estimated to require a roughly equal mix of physical, cognitive, and interpersonal time.

Appendix Figure 1 plots tasks by time spent on physical, cognitive, and interpersonal work. Work time is split into 5% units adding to 100%, so task locations measure time demands. Circle sizes show total working time in the US economy. Each task is weighted by the number of workers who do it, and the fraction of working time they spend on it. Employment data come from the Occupational Employment and Wage Statistics (BLS, 2025), and we estimate task time demands with Claude. Tasks in the center require all three types of work. For example, “Teach dance students” requires 35% physical, 20% cognitive, and 45% interpersonal time. Other work, like “Dig trenches,” is mostly physical (85%), though this task requires some judgment and team coordination.

Tasks we identify as physical are shaded orange. These require at least 50% physical work time, require physical skills that take months or more to build, or could not be accomplished without physical work. Specifically, for task t we define

$$physical_t = 1[time_{tp} \geq 50 \text{ or } level_{tp} \geq 3 \text{ or } essential_{tp} = 1]$$

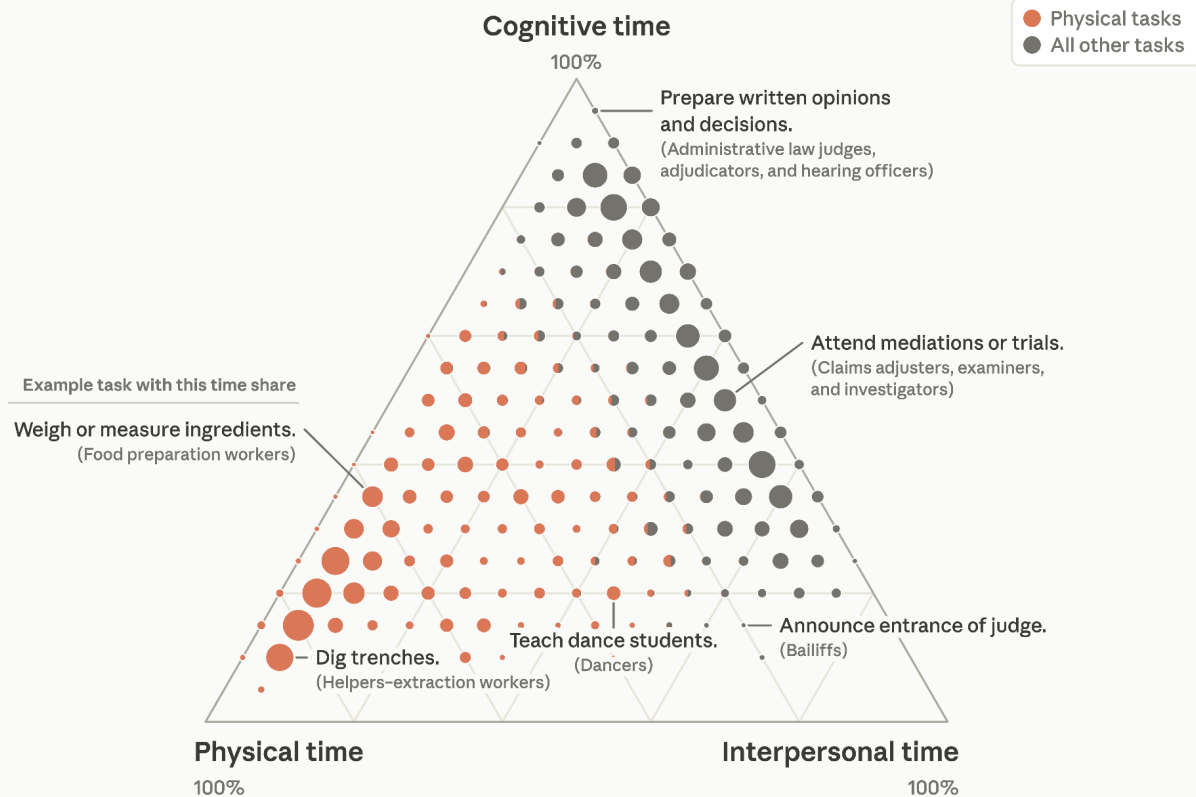
“Dig trenches” meets all three criteria. “Teach dance students” also counts despite being only 35% physical by time, since it requires physical expertise built over years and could not be accomplished without physical skill.

As shown in Appendix Figure 1, tasks cluster in the bottom-left physical corner, and along the cognitive-interpersonal edge on the right. Most work is either mostly physical, or a mix of cognitive and interpersonal. The fact that many physical tasks do not also require much cognitive or interpersonal input suggests that a physically capable robot could perform most physical tasks in the economy. We assess robot exposure for the roughly 7,600 tasks classified as physical.

In this figure and throughout, we drop occupations without O*NET task descriptions and those without employment and wage data from BLS (2025). These jobs account for 3% of employment and are more likely to be in the two broad occupation groups “Management, business, science & arts” and “Production, transportation & material moving.” Since the former is likely less exposed to robots and the latter more, these exclusions on net seem unlikely to change overall exposure estimates.

Job tasks by time spent

On physical, cognitive, and interpersonal work



Appendix Figure 1: Job tasks by time spent on physical, cognitive, and interpersonal work

Notes: Tasks are rated on physical, cognitive, and interpersonal time demands in units of 5% adding to 100%. Circles group tasks by time shares, and circle areas are proportional to the estimated share of all working time spent on those tasks. We calculate this by estimating how much time an occupation spends on each of its tasks using Claude and multiplying by occupation employment. For example, “Teach dance students” requires 35% physical, 20% cognitive, and 45% interpersonal time. Circles are shaded in proportion to time on tasks we identify as physical. Employment data come from BLS (2025).

A.2 Task taxonomy validation

To validate these ratings, we link occupation time and skill level variables to data from the 2023 and 2025 BLS Occupational Requirements Surveys, a survey of employers to collect data on job requirements. We identify a subset of variables describing physical, cognitive, and interpersonal job requirements.¹ These are aggregated into indices for physical demands, problem solving skills, and social demands.²

We compare these occupation requirements to our own scores of how physical, cognitive, and interpersonal jobs are. We average task time and skill variables

by occupation, weighting by time spent on tasks. For each occupation o and each component $k \in \{physical, cognitive, interpersonal\}$, define

$$time_{ok} = \sum_{t \in o} (time_{tk}/100) w_t$$

$$level_{ok} = \sum_{t \in o} level_{tk} w_t$$

Here, w_t is our estimate of the share of time occupation o spends on task t . Variable $time_{ok}$ therefore measures share of time spent on physical, cognitive, and interpersonal work, while $level_{ok}$ measures time-weighted skill demands.

Appendix Figure 2 displays correlations between our ratings and the BLS occupational requirement indices. The corresponding sets of physical, cognitive, and interpersonal variables are quite correlated. For example, the physical time share correlates 0.76 with the physical demands index, and the cognitive and interpersonal skill levels correlate 0.89 and 0.78 with the problem solving and social demands indices. All physical variables negatively correlate with all cognitive and interpersonal variables. Meanwhile, cognitive and interpersonal variables are positively correlated. These patterns mirror those in Appendix Figure 1.

A.3 Robot exposure

Our occupation robot exposure index begins by rating exposure for tasks classified as physical, as described above. For these tasks t , we measure ordinal task $exposure_t \in \{E0, E1, E2, E3\}$ by the kind of environment E that a robot requires to perform this task:

- E0: Robot cannot perform task.
- E1: Robot can perform task in a purpose-built robotic work environment, like a factory assembly line.
- E2: Robot can perform task in a structured human work facility, like a logistics warehouse.
- E3: Robot can perform task in an unstructured environment, like a city road.

We say that a robot can perform a task if it produces a unit of output at quality an employer would accept. To focus on potential job automation, our prompt emphasizes that a robot must act largely without human control. That

Time and skill ratings against BLS job requirements

Occupation ratings		Correlation with BLS occupational requirements		
		Physical demands	Problem solving	Social demands
Physical	Time share	0.76	-0.67	-0.72
	Skill level	0.80	-0.54	-0.56
Cognitive	Time share	-0.69	0.65	0.43
	Skill level	-0.43	0.89	0.47
Interpersonal	Time share	-0.55	0.44	0.83
	Skill level	-0.53	0.73	0.78
N (occupations)		546	442	571

Appendix Figure 2: Time and skill ratings against BLS job requirements

Notes: Job time and skill ratings, displayed in rows for physical, cognitive, and interpersonal work, are calculated using estimated task time and skill ratings. Physical demands, problem solving, and social demands are indices calculated using data from the 2023 and 2025 BLS Occupational Requirements Surveys. For example, cognitive skill requirements in our taxonomy are highly correlated with BLS job requirements for problem solving skills (0.89). Entries are occupation-level correlations weighted by employment. Occupations are recorded by 2019 O*NET-SOC codes.

excludes, for example, teleoperated surgical machines which humans control. See Figure 2 and Appendix F.3.

We prompt Claude to measure $exposure_t$ using web search. To establish exposure, Claude is required to name a specific robot and provide direct source quotes to justify its claims. Including historical exposure ratings (described below), Claude makes around 650,000 web searches and cites over 275,000 quotes from over 90,000 unique web pages.

Before rating task exposure, we prompt Claude to describe representative instances of a task and how often they occur. This step aims to elicit details about physical work that may be easy for humans but hard for robots, as described through the “Dig trenches” example in the main text. Claude rates exposure for each task instance. Task exposure is the highest exposure level for which at least half of instances by weight are exposed at that level or higher. Specifically,

$$exposure_t = \max\{e \in \{0,1,2,3\} : \sum_{i \in t} 1[exposure_{it} \geq e] \cdot w_{it} \geq 0.5\}$$

Here, $1[exposure_{it} \geq e]$ indicates whether exposure for task instance i of task t is rated exposure e or higher, and w_{it} is the weight for instance i in task t .

Aggregating tasks, an occupation's *robot exposure index* is defined by

$$exposure_o = \sum_{t \in o} exposure_t \cdot w_t$$

This index ranges from 0 to 3, and $exposure_t$ is set to 0 for unrated tasks.

A.4 Alternative robot exposure measures

We also consider several other ways of measuring exposure. These produce similar conclusions about robot exposure.

Our exposure classifier counts robot capabilities shown in demonstrations when Claude finds these credible. Demonstrations aren't as solid evidence that robots can do tasks as commercial deployments. But dropping tasks that cite only demonstrations, which we explicitly recorded as part of the evidence retrieval, lowers the share of exposed physical work by only 1 percentage point. It also produces nearly the same ranking of jobs by exposure as our main measure (rank correlation 0.99).

Sometimes Claude can't find a robot that does a task exactly as written, but counts the task as exposed when robots do related ones. For example, automotive service technicians and mechanics "Align wheels, axles, frames, torsion bars, and steering mechanisms of automobiles, using special alignment equipment and wheel-balancing machines." Claude can't find a robot that does so in repair shops, where this occupation typically works, but rates this task E1 because of wheel-aligning robots in car factories.³ For "Dig holes for power poles, using power augers or shovels, set poles in place with cranes, and hoist poles upright, using winches," Claude cites a pile-driving robot that moves around solar farm sites and drives steel poles into the ground, along with other robots that dig.⁴

Excluding ratings that rely on related robots decreases the share of exposed physical work from about three-quarters to a half. Though these cases require judging whether capabilities transfer across tasks, they show which tasks robots could likely do, or are close to doing, with today's technology. We also

define an exposure measure based on the gap between where humans perform a task and where robots can do it today:

$$exposureCoverage_o = \sum_{t \in o} (exposure_t / humanEnvironment_t) \cdot w_t$$

Here, $humanEnvironment_t$ is the typical human working environment for task t . Taking the ratio between robot exposure and human working environments counts a task less if robots are further from doing it like humans. This occupation exposure index ranges from 0, meaning robots cannot do any tasks, to 1, meaning robots can do all tasks in the same environments as do workers. Again, this measure is highly rank-correlated with our primary exposure index, at 0.99.

In Figure 6, we combine robot exposure with LLM exposure measures from Eloundou et al. (2024). Specifically, we define

$$llmExposure_o = \sum_{t \in o} 1[\beta_t > 0] \cdot w_t$$

$$llmAndRobotExposure_o = \sum_{t \in o} 1[\beta_t > 0 \text{ or } exposure_t > 0] \cdot w_t$$

Here, β_t is Eloundou et al.'s GPT-4 task exposure measure. Eloundou et al. use an older O*NET edition, so these calculations omit around 50 tasks.

A variant of this measure that counts fractional rather than binary robot exposure for each task produces a similar pattern of results. Specifically, we calculate the share of a task's instances that are exposed, and use that to measure LLM and robot exposure:

$$exposureFrac_t = \sum_{it \in t} 1[exposure_{it} > 0] \cdot w_{it}$$

$$llmAndRobotExposureFrac_o = \sum_{t \in o} \max\{1[\beta_t > 0], exposureFrac_t\} \cdot w_t$$

For example, exposure in production jobs decreases from around 90% to 80% when using the share of exposed task instances. This may reflect instances where today's robots encounter complex exceptions that require humans to handle. For example, 75% of instances are rated E1 for the task "Spray chopped fiberglass, resins, and catalysts onto prepared molds or dies using pneumatic spray guns with chopper attachments." Robots can spray fiberglass on molds for large structures like yacht hulls, but Claude notes that they can't yet do the task instance, "Complex molds and detail work: spraying deep narrow recesses, lockers, undercuts and flange returns, and spraying over foam cores,

rib/stiffener formers, inserts and hardware,” and estimates this instance takes 15% of task time.⁵

B. Historical robot exposure

Our measures of job exposure to robots today are most informative if they predict which jobs are more or less likely to face disruption in the future. We validate our measure by connecting historical robot exposure to changes in job employment and pay over the past 50 years.

B.1 Historical task and labor market data

To track robot exposure and labor market outcomes over time, we compile task description and labor market data for 5 periods from 1977 to today. Sources are listed in Appendix Figure 3. We take snapshots of the US labor market decadal from 1980 to now, taking employment, wage, demographic, and occupation data from the 5 percent Census when available and the 5-year American Community Survey when not.⁶ From the historical O*NET and DOT editions close to these dates, we get rich descriptions of work as it existed at the time.⁷

Using DOT data, we construct job titles, descriptions, and task statements to roughly match the structure of O*NET. DOT data enumerate about 12,000 jobs with paragraphs describing specific work duties. Similar to Kogan et al. (2023) and Autor and Thompson (2025), we split sentences into task descriptions.

Our labor market data cleaning follows Autor et al. (2024). To link job descriptions and labor market data over time, we crosswalk DOT occupation codes, O*NET-SOC codes, and Census occupation codes following Autor and Dorn (2013) and Autor et al. (2024). Census occupation codes identify around 300 consistently defined jobs in our sample. When aggregating finer occupation codes to linked Census codes, we weight by estimated employment in that period.⁸ To compute average occupation wages, we first adjust individual wages each decade by race, sex, education, and age (Autor et al., 2024). This adjustment helps remove effects of changes in worker composition in an occupation over time.⁹ Employment outcomes account for hours worked and so measure full-time-equivalent workers. We use Census rather than OEWS data, which we use in present-day analyses, to keep a consistent labor

Historical data sources by decade

Decade	Task description data	Labor market data
2020s	O*NET 29.3 (2025)	American Community Survey (2020–2024)
2010s	O*NET 13.0 (2008)	American Community Survey (2008–2012)
2000s	O*NET 4.0 (2002)	Census (2000)
1990s	DOT, revised 4th edition (1991)	Census (1990)
1980s	DOT, 4th edition (1977)	Census (1980)

Appendix Figure 3: Historical data sources by decade

Notes: Rows list data sources for job task descriptions and labor market statistics by decade.

market dataset and because the latter isn’t available before 1997. We comment on potential differences between the two as it pertains to our results later.

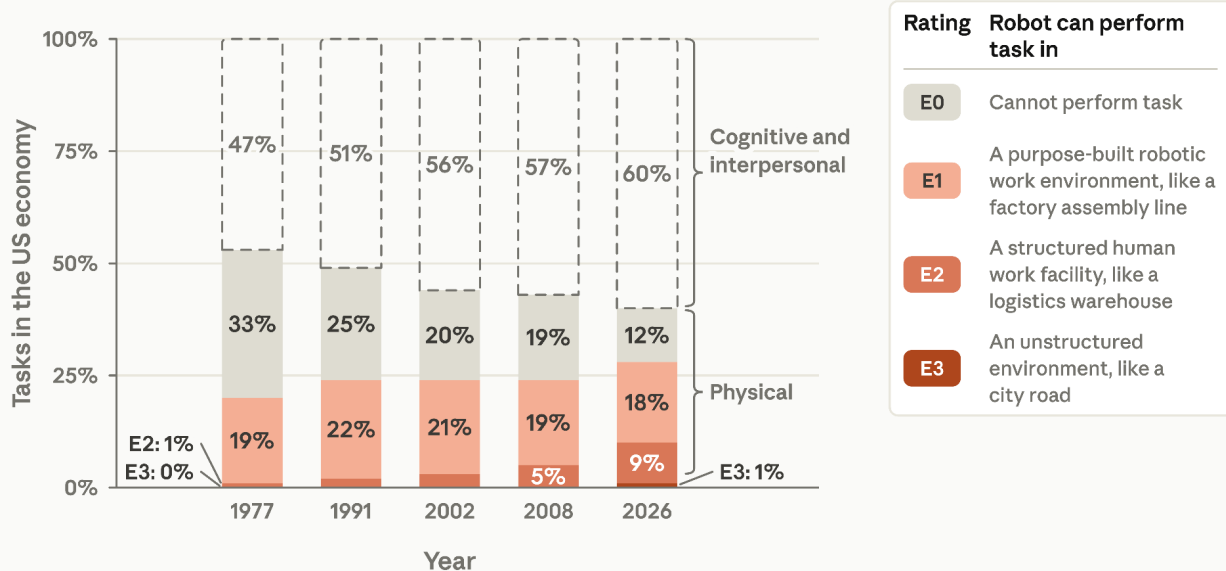
The result is a roughly decadal panel of jobs by occupation and industry. Employment, wages, and demographics vary across occupation by industry cells, and historical robot exposure is measured for occupations.

B.2 Measuring historical robot exposure

Ideally, we would measure historical robot exposure with an equally capable version of Claude trained only on historical data. Models trained on modern data “know” which tasks in 1977 were eventually automated, which could contaminate historical judgments.¹⁰ To mitigate this look-ahead bias, we prompt Claude to consider work as it existed in the target year and require sources that place any cited robot in the appropriate time.¹¹ Historical robot exposure is constructed as described above using these judgments.

Appendix Figure 4 plots working time in the US economy by robot exposure in each period from 1977 to 2026, weighting tasks by estimated work time requirements and Census employment. Large exposed occupations in 1977 included textile sewing machine operators and assemblers of electrical equipment.¹² The trend in this figure captures a novel, albeit expected, result

Job tasks by robot exposure over time



Appendix Figure 4: Job tasks by robot exposure over time

Notes: Job tasks come from the DOT (for 1977, 1991) and O*NET (for 2002, 2008, 2026). Robot exposure is rated for tasks identified as physical. A task is weighted by the estimated fraction of time it requires in an occupation and by occupation employment. Employment data come from the Census (1980, 1990, 2000) and the American Community Survey (2008-2012, 2020-2024). For example, in 2002, 56% of job tasks by working time were cognitive and interpersonal; robots could do 21% of tasks in purpose-built robotic work environments.

about the economy: Work in the US has become less physical over time, falling from 53% of job tasks to 40% today. In part, this captures the decline in US manufacturing over this period. That 13 percentage point drop represents a 25% decline in the share of physical work since 1977.

For reference, Atalay et al. (2020) use changes in words from newspaper job ads between 1950 and 2000 to study changes in work. Weighted by Census employment, they find the frequency of words describing routine manual work (e.g., “operating”) declined 94% and nonroutine manual work (e.g., “renovating”) 18%. Our estimates aren’t directly comparable, but they likewise point to a decline in physical work. Atalay et al. also find that routine cognitive words (e.g., “bookkeeping”) became less frequent; that could partially offset the large routine manual declines and explain our more modest estimated decline in the share of physical work overall.¹³

Weighting by Census rather than OEWS employment lowers the estimated share of physical work today: 40% here, and 46% in Figure 3. One reason could

be differences in how workers and employers describe jobs. Abraham and Spletzer (2010) find that the CPS, another worker survey, consistently counts more management jobs and fewer service jobs than the OEWS. That could explain the level difference between Census and OEWS estimates of physical work; trends are nearly identical when calculated with OEWS data since 2002.¹⁴

As the share of physical work has shrunk, the share that robots can perform has risen. In 1977 around 62% of physical tasks could not be performed by any robot. That figure today is 30%. This comparison mixes changing robot capabilities with the changing nature of work. Appendix B.4 presents estimates that isolate the trend in robots.

Exposed tasks and associated jobs also show how robots have advanced over time. For example, all tasks for production line welders in 1977 are rated at E1 exposure, reflecting the robotic welding arms used in auto body factory lines. One notable pattern in Appendix Figure 4 is the recent rise in E2 and E3 robot exposure. An important characteristic of these robots is that they can often move around rather than needing to be fed inputs at a fixed location. While workers that move materials around were E1 exposed in 1977 due to palletizers, mobile warehouse robots now expose such workers at level E2.

B.3 Historical robot exposure and labor market impacts

We connect what robots could do to labor market impacts, tracking exposure and job data in three overlapping periods of roughly 20 years. We regress changes in log wages and employment on starting year robot exposure, pooling data across these three periods from 1977 to 2024.

Since past robots mostly worked in structured environments like factories, we measure historical job exposure by the time-weighted share of tasks exposed at E1 or above. Historical exposure is related to labor market outcomes with regressions of the form:

$$\Delta \ln y_{oip} = \alpha \cdot \text{exposedTaskTime}_{op} + \gamma_{ip} + \varepsilon_{oip}$$

Here, $\Delta \ln y_{oip}$ is the change in occupation o 's adjusted log wage or employment in industry i , starting from period p , and $\text{exposedTaskTime}_{op}$ is the share of occupation o 's tasks that robots could perform in the beginning of period p . Of

course, the US economy underwent large changes over this period not attributable to robots, like trade with China and computerization. Regressions therefore compare changes in employment and wages for jobs more or less exposed to robots in the same industry, operationalized by adding industry by period fixed effects γ_{ip} .

The coefficient α gives the approximate percent change in wages or employment over 20 years for an occupation with all tasks exposed to robots, relative to an unexposed occupation in the same industry. We pool the three periods and weight observations by initial employment, normalizing so that each period counts equally. Standard errors are clustered by occupation.

Appendix Figure 5 presents results. The left panel plots occupations by robot exposure and changes in log wages averaged over the three periods. Compared to an unexposed occupation in the same industry, wages were 7.1% lower 20 years later for an occupation whose tasks could all be done by robots. This effect is precisely estimated with a 95% confidence interval of 5.4% to 8.9%. For example, welders, solderers, and metal cutters had three-quarters of their work time exposed to robots on average. Over the next 20 years, their wages grew about 3% less than for workers in other occupations in the same industries.¹⁵

Employment also fell for exposed occupations (right panel). Relative to an unexposed occupation in the same industry, an occupation whose tasks can all be done by robots is predicted to lose 34.2% of its jobs over about 20 years, with a 95% confidence interval of 16.4% to 52.0%.

The auto industry, where robots are most common and are widely believed to have eliminated jobs, shows the pattern.¹⁶ In 1991, machine operators were in the top 5% of all workers by robot exposure, because robots such as automated painters and welders could do the same work. About 100,000 such workers held jobs in the motor vehicle sector in 1990. By 2012, about 50,000 did. The sector shrank overall, but not all its workers lost jobs. Industrial machinery repairers, whose exposure was one-quarter that of machine operators, saw a 50% rise in motor vehicle employment. Production supervisors or foremen, who were nearly unexposed, saw a 10% employment rise.

Wage and employment declines fit with other work on robots in the US, though findings in other countries are mixed. As the case of motor vehicles illustrates,

Labor market effects of robot exposure over time



Appendix Figure 5: Labor market effects of robot exposure over time

Notes: The lines plot the slopes from regressions of log wage and employment changes on historical robot exposure, weighted by initial employment and with industry by period fixed effects. Estimates pool data from three overlapping periods of roughly 20 years from 1977 to 2024. Wages and employment are adjusted for industry trends.

industrial robots may have displaced production workers but supported supervisors and repairers, consistent with results from other studies.¹⁷

We next explore how robot effects vary over time. Appendix Figure 6 reports results, with main regression estimates in column 1 and estimates separately by period in later columns. Across the three periods of about 20 years shown in columns 2-4 of the top panel, wages consistently decline around 7% for fully exposed occupations. Though regressions include industry fixed effects, it's possible that steady wage declines reflect secular trends like the growth in services rather than robots. If so, robot exposure might predict occupation wage changes even before robots were adopted. A placebo test in column 5 shows this is not the case: robot exposure as of 1977 has no effect on wages from 1970 to 1980.

Employment effects appear in the bottom panel. Unlike wages, employment effects seem to attenuate over time. From 1980-2000, employment fell around

Historical exposure regressions by period

	Exposure effects by period				Placebo effects
	1980–2024	1980–2000	1990–2012	2000–2024	1970–1980
	(1)	(2)	(3)	(4)	(5)
Change in log adjusted wage					
Robot exposed time share	–0.071 (0.009)	–0.052 (0.013)	–0.074 (0.009)	–0.083 (0.012)	–0.004 (0.010)
N (occupations × industries)	83,856	28,143	27,999	27,714	20,176
N (occupations)	305	305	304	301	245
Change in log employment					
Robot exposed time share	–0.342 (0.091)	–0.517 (0.165)	–0.275 (0.102)	–0.279 (0.140)	0.028 (0.215)
N (occupations × industries)	85,192	28,581	28,393	28,218	20,499
N (occupations)	306	306	305	302	246

Appendix Figure 6: Historical exposure regressions by period

Notes: The top panel reports estimates for wages and the bottom for employment. Columns report estimates by period. Regressions control for industry fixed effects and weight by initial employment. Exposure is as of 1977 for the period 1980–2000, as of 1991 for 1990–2012, and as of 2002 for 2000–2024. Column 1 pools these three periods and weights them equally. Column 5 uses exposure as of 1977 for the period 1970–1980 as a placebo test. Standard errors, clustered by occupation, are in parentheses.

50% for exposed compared to unexposed occupations. This estimate is about half as large in 1990–2012 and 2000–2024 but remains significant.

Wage and employment results may diverge because new types of jobs, like service jobs, became exposed over time. In 1977, most exposed jobs were highly physical: among those with more than a third of work time exposed to robots, around half spent at least 95% of time on physical tasks. Only around 24% of such jobs spent as much time on physical tasks in 2002. Robots seem more likely to displace jobs that are highly physical compared to those that also involve cognitive or interpersonal work, so broader exposure may have muted employment declines while maintaining wage effects.

As with wages, column 5 shows that exposure as of 1977 doesn't predict employment changes from 1970 to 1980, though this estimate is imprecise.

Though the 1970–1980 placebo tests suggest that regressions isolate robots from other trends, Appendix Figure 7 stress tests our results with several alternative explanations. Column 1 reproduces our main results, and columns 2–4 show how results change when adding control variables.¹⁸ Column 2 compares our robot exposure measure to that of Webb (2020), which defines exposure by text similarity between O*NET descriptions and robotics patents. Like this paper, Webb (2020) shows that robot exposure predicts wage and employment declines. Regressions that include both exposure ratings find larger effects for our measure based on demonstrated robot capabilities, which makes sense since capabilities are downstream of patents.¹⁹

Robot exposure remains a strong predictor of labor market impacts after adjusting for skill-biased technological change, offshoring, and deunionization. Robots spread alongside other technologies, notably computers, which are thought to have raised demand for skilled labor and automated routine tasks.²⁰ For example, computer software made obsolete the keypunch operator's task “Depresses keys to transcribe new data in prescribed sequence from source material into perforations on card, or as magnetic impulses on specified locations on tape or card.” Our measure excludes this task from robot exposure: Claude judges that it aims to transmit information, not act on physical objects. Other technologies would confound our results only if they affect physical tasks specifically.

Consistent with this, estimated effects of robot exposure are robust to adjustments for several potential confounders. Column 3 reports regressions that control for variables often used as skill proxies: baseline wages, education, age, sex, national origin, and race.²¹ Effects attenuate somewhat and remain strongly significant for wage declines; the 95% confidence interval for employment narrowly includes zero. Additional controls for changes in unionization and for job offshorability, from Autor and Dorn (2013), produce similar results, and employment effects return to statistical significance.²²

Finally, we benchmark our estimates against Webb (2020). For manufacturing jobs from 1980 to 2010, Webb (2020) finds that an occupation at the 75th percentile of exposure had up to 18% lower employment and 14% lower wages than one at the 25th percentile. Split over three decades, that amounts to about 6% and 5% per decade for employment and wages. For a rough comparison, we multiply the period-specific coefficients in Appendix Figure 6 by the

Exposure regression robustness checks

	<u>Main results</u>	<u>Patent exposure</u>	<u>Baseline demos & wages</u>	<u>Offshoring and unions</u>
	(1)	(2)	(3)	(4)
Change in log adjusted wage				
Robot exposed time share	-0.071 (0.009)	-0.083 (0.011)	-0.051 (0.010)	-0.053 (0.010)
Patent robot exposure		0.005 (0.003)		
Offshorability				-0.002 (0.003)
Union density change				-0.001 (0.002)
Baseline demos & wages			✓	✓
N (occupations × industries)	83,856	83,566	83,856	83,856
N (occupations)	305	302	305	305
Change in log employment				
Robot exposed time share	-0.342 (0.091)	-0.291 (0.112)	-0.267 (0.139)	-0.305 (0.152)
Patent robot exposure		-0.024 (0.031)		
Offshorability				0.000 (0.024)
Union density change				-0.030 (0.028)
Baseline demos & wages			✓	✓
N (occupations × industries)	85,192	84,897	84,464	84,464
N (occupations)	306	303	306	306

Appendix Figure 7: Exposure regression robustness checks

Notes: Column 1 reproduces main results. Column 2 adds patent robot exposure from Webb (2020). Column 3 controls for baseline occupation by industry log wage and demographic shares calculated from the Census: age (5 bins), sex (2), race (4), foreign born (2), and education (4). Column 4 adds occupation offshorability from Autor and Dorn (2013) and the change in union membership rate over the period from Hirsch et al. (2026) and the CPS. Patent exposure, offshorability, and union variables are standardized across occupations weighting by employment. Patent exposure and offshorability are defined at a finer occupation level and aggregated to our codes using employment weights, which affects around 10% of occupations.

interquartile range of exposed task time in 1980 and 1990, and halve them to get decadal effects. Moving from the 25th to the 75th percentile of our exposure measure predicts a 9% employment decline per decade for 1980–2000 and 6% for 1990–2012. Our larger but comparable employment estimates are consistent with the stronger predictive power of capability-based exposure measures noted in Appendix Figure 7. Effects for wages are also comparable.²³

Overall, these results indicate that our robot exposure measure can help identify occupations at risk of labor market impacts from physical automation. A caveat is that robots in this era were mostly used in controlled, E1 factory settings, so historical data say little about the E2 or E3 exposure that more adaptable, AI-controlled robots may create. Robots that work with less workplace engineering could have larger effects on jobs.

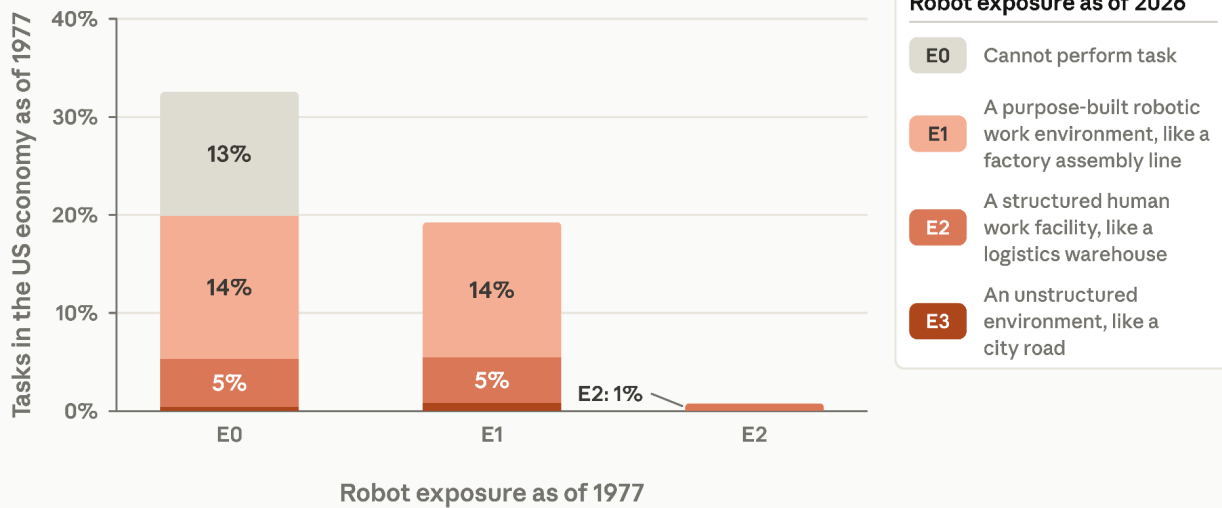
B.4 How quickly have robot capabilities advanced?

New robots will be able to do new tasks that they can't today. To measure the historical rate of progress, we rate job tasks from 1977 using robots as of 2026. Fixing tasks as of 1977 isolates how fast new robots became able to do work they previously could not do.

Appendix Figure 8 plots exposure today for job tasks in 1977.²⁴ Robots can now do around 60% of the tasks they could not before.²⁵ Robots that help herd, milk, feed, and vaccinate cows can monitor health and administer medicine, so the 1977 task “Cares for livestock and poultry, observing general condition and administering simple medications to animals and fowls” is now rated E2.²⁶ On construction sites, a bricklaying robot now “Spreads soft bed of mortar that serves as base and binder for block, using trowel.”²⁷ And an autonomous mining vehicle “Drives machine into pile of material blasted from working face” to scoop ore.²⁸ From 1977 to today, robots became able to perform about 2% more of the physical work they previously could not per year.²⁹

Robots also improved at tasks they could already do. In 1977, automated photoelectric sorters could detect unripe produce on a conveyor belt and eject it automatically. Because of this and other systems, “May examine and sort items tallied” for talliers (clerical professionals who account for inventory) was rated E1 in 1977, with just over half of task instances exposed.³⁰ Today this task is E2: a warehouse robot with computer vision and suction grippers can sort

Robot exposure today for 1977 job tasks



Appendix Figure 8: Robot exposure today for 1977 job tasks

Notes: Bars group 1977 job tasks by exposure to robots as of 1977. Segments in each bar show exposure for these same tasks to robots as of 2026. For example, of E1 tasks in 1977, a quarter (5% / 20%) are E2 when considering robots in 2026. Tasks rated E3 account for less than 0.05% of 1977 working time and are omitted. Tasks are weighted by estimated working time as of 1977 and employment from the 1980 Census.

packages and pick products of many different kinds.³¹ Higher exposure reflects better abilities, so each year since 1977 robots became more adept at doing 0.7% of the tasks they previously could do only in purpose-built settings.

These ratings also show that physical automation goes beyond robots. Transcribing-machine operators used to do the task “Turns dials to control volume, tone, and speed of voice reproduction.” Robots can now manipulate simple dials, but they hardly matter for this task compared to digital audio and speech recognition.

Some tasks still resist automation, like a furniture upholsterer’s “Removes covering, webbing, and padding from seat, arms, back, and sides of workpiece, using tack puller, chisel, and mallet.” Fabrics and deformable objects remain hard for robots, as in 1977. Other technologies improve physical work, but do not count as robots in our framework because a human operates them. Surgeons today use the da Vinci system for urological, colorectal, and thoracic operations. Often called a surgery robot, we do not count it since a surgeon operates it with a controller that transmits finger and hand movements to

mechanical arms.³² For other tasks, robots have improved but not enough. Robots are just now becoming able to route wires around pins on flat, open boards in factories. But they lack the dexterity to do the task “Connects electrical wires on control panel to motors, heating units, or blowers, according to color and length of wire, by forcing plug connectors together and sliding plastic insulators over connection.”

Nearly the same share of job tasks in 1977 and 2026 are unexposed to today’s robots, 13% here versus 12% in Appendix Figure 4. Work has changed over time, but jobs and tasks have shifted in offsetting ways. Farming, mining, production, and operation declined as a share of work, while personal services and healthcare grew. Tasks in both groups involve hands-on work, so the share of unexposed work remains about the same. Fewer workers do tasks like “Resews garment, using needle and thread or sewing machine,” while more do work like “Provide patients with help moving in and out of beds, baths, wheelchairs, or automobiles and with dressing and grooming.”

C. Barriers to automation

C.1 What work is not exposed to robots or LLMs?

As Figure 6 shows, most tasks are exposed to robots or LLMs, but 19% are not. Here we analyze unexposed task statements to characterize this work and identify possible barriers to automation.

We use Claude to interpret unexposed tasks. First, we partition them into 10 groups using k-means on task statement embeddings. Second, we ask Claude to describe each group based on a sample of 100 tasks. For contrast, Claude also receives the 100 unexposed tasks closest but not belonging to it, and the same for the 100 closest exposed tasks.³³

Appendix Figure 9 describes unexposed work, 56% of which is physical. The largest category, “Directing, training, and coordinating workers,” makes up 2.4% of all US tasks by working time. Recreation workers, chefs and head cooks, and construction managers, among others, perform these tasks. Managing workers is interpersonal, but 12% of these tasks are also classified as

Tasks not exposed to robots or LLMs

Description of unexposed task group	Share of all job tasks	Physical tasks in group	Example task
Directing, training, and coordinating workers	2.4%	12%	Supervise and coordinate the work activities of personnel, such as training staff members and assigning work duties. (Recreation workers)
Personal care, dressing and escort assistance	2.3%	81%	Provide patients with help moving in and out of beds, baths, wheelchairs, or automobiles and with dressing and grooming. (Home health aides)
Hands-on instruction and student supervision	2.3%	42%	Instruct students individually and in groups, using various teaching methods, such as lectures, discussions, and demonstrations. (Self-enrichment teachers)
Diagnosis and repair of vehicles and equipment	2.1%	89%	Inspect, clean, maintain, and repair equipment, using mechanics' hand tools, or report malfunctions to supervisors. (Paving, surfacing, and tamping equipment operators)
Hands-on clinical procedures and treatments	2.0%	85%	Administer medications to patients and monitor patients for reactions or side effects. (Registered nurses)
Housekeeping and personal grooming services	1.7%	97%	Perform or assist with cleaning duties as necessary. (First-line supervisors of housekeeping and janitorial workers)
Face-to-face patron assistance and selling	1.6%	11%	Greet customers and ascertain what each customer wants or needs. (Retail salespersons)
Building systems installation and construction	1.5%	87%	Repair machines, equipment, or structures, using tools such as hammers, hoists, saws, drills, wrenches, or equipment such as precision measuring instruments or electrical or electronic testing devices. (Maintenance and repair workers, general)
Rule enforcement and public safety protection	1.5%	45%	Monitor and authorize entrance and departure of employees, visitors, and other persons to guard against theft and maintain security of premises. (Security guards)
Client counseling and case conferring	1.4%	4%	Counsel individuals, groups, families, or communities regarding issues including mental health, poverty, unemployment, substance abuse, physical abuse, rehabilitation, social adjustment, child care, or medical care. (Child, family, and school social workers)
All tasks unexposed to robots or LLMs	18.8%	56%	

Appendix Figure 9: Tasks not exposed to robots or LLMs

Notes: Tasks unexposed to robots (E0 or not physical) or LLMs (zero β from Eloundou et al. (2024)) are clustered into 10 groups. Clusters come from k-means on task description embeddings, weighted by task time and employment. Descriptions are written by Claude using a sample of tasks in the group and the nearest tasks outside it by embedding distance. Example tasks are those closest to the center of each cluster. The 2nd row shows, for example, that the “Personal care, dressing and escort assistance” group covers 2.3% of all job tasks, and 81% of them are identified as physical tasks. Its example task is moving patients in and out of bed.

physical, like “Demonstrate new cooking techniques or equipment to staff” for chefs and head cooks.

More broadly, managing and teaching often involve showing by physical example. As with cooking techniques, “Hands-on instruction and student supervision” involves physical learning, like demonstrating self-enrichment

activities to students. Robots that cook or paint would be useful for these tasks only if cooks or students learned as well from watching robots as from humans: a robot chef with suction-cup grippers may not help a human learn to fillet a fish. Humans may also respond better to human compared to AI instructors.

That said, some management and training may not be needed if robots and AI automate workers themselves. Robots today cannot cook like humans, but if they could, training chefs might matter less. More immediately, observers have raised concern that AI will automate entry-level office jobs.³⁴ That could eliminate the task “Supervise, train, or mentor junior team members” for financial and investment analysts. Automation seems less likely to reduce demand for teaching children than for teaching workers.

Likewise, demand for “Face-to-face patron assistance and selling” tasks may arise mostly from human interaction rather than assistance per se, so tasks like “Greet customers and ascertain what each customer wants or needs” may not be automated even if technically possible.

Two groups of physical tasks seem mainly limited by robot capabilities: “Diagnosis and repair of vehicles and equipment” and “Building systems installation and construction.” Consider a task in the latter group for carpenters: “Install structures or fixtures, such as windows, frames, floorings, trim, or hardware, using carpenters’ hand or power tools.” To do so, carpenters must exercise many physical abilities in unstructured environments. By O*NET ability requirements, carpenters rank in the top 5% of workers on manual dexterity, arm-hand steadiness, finger dexterity, and dynamic strength, among other abilities.³⁵ No robot today has the abilities to perform that task.

More capable robots could also perform other physical tasks, though preferences for humans could remain important here. Safely touching humans is difficult, so robots today do not “Provide patients with help moving in and out of beds, baths, wheelchairs, or automobiles and with dressing and grooming.” Patients may also prefer human home health aides to capable robot ones. Similar preference and capability factors could slow automation for some tasks in “Housekeeping and personal grooming services,” such as “Cut and trim hair according to clients’ instructions or current hairstyles, using clippers, combs, hand-held blow driers, and scissors.” By contrast, robot vacuums and demonstrations of humanoids doing household chores suggest greater appetite for robot housekeepers.

Unexposed tasks in medicine and law also highlight regulation as a barrier to automation. Regulation seems especially important for “Hands-on clinical procedures and treatments,” “Client counseling and case conferring,” and “Rule enforcement and public safety protection,” which together account for 5% of tasks. For example, robots that “Administer medications to patients and monitor patients for reactions or side effects” or “Identify, pursue, and arrest suspects and perpetrators of criminal acts” would likely face legal challenges. Rules may change as technology develops, but could still slow automation.

Overall, tasks unexposed today are highly interpersonal and physical. Their descriptions suggest that human preferences, missing physical capabilities, and regulation could keep robots and LLMs from doing them. We next explore these possible barriers to automation in more detail.

C.2 Barriers to automation

Our exposure measure ranks jobs by what today’s robots can do, but we aim to understand how the economy will change in the future. To automate work, robots must be capable but also overcome other barriers. Task descriptions in Appendix C.1 point to regulation and human preferences, and robot costs matter as well. Here we take a first look at barriers to automation for physical tasks. We include exposed and unexposed tasks, since robots haven’t automated either at scale.

For each physical task, we ask Claude to identify barriers to automation today that, on their own, would prevent robots from doing a significant share of this work. Barriers include capability, cost, human preference, and regulation. For capability, Claude picks the most important barrier among four robot skills: manipulation, planning and reasoning, mobility and strength, and perception.

We instruct Claude to cite sources for its judgments and name specific regulations. It also receives the output from our robot exposure classifier. For tasks that robots today do not do, Claude judges whether a robot could be built with today’s technology. If not, capability is a barrier. If so, Claude judges what would be a barrier if one existed, drawing on similar robots today. These estimates are more forward-looking than our robot exposure measure, at the cost of being less grounded in today’s robots.³⁶

Appendix Figure 10 shows barriers to automating physical tasks. Physical work is too expensive to automate today: cost is a barrier for nearly every task. That makes sense, since tasks describe what humans do today. As with other technologies, robot hardware costs are likely to fall over time. We consider costs in more detail in Appendix D.

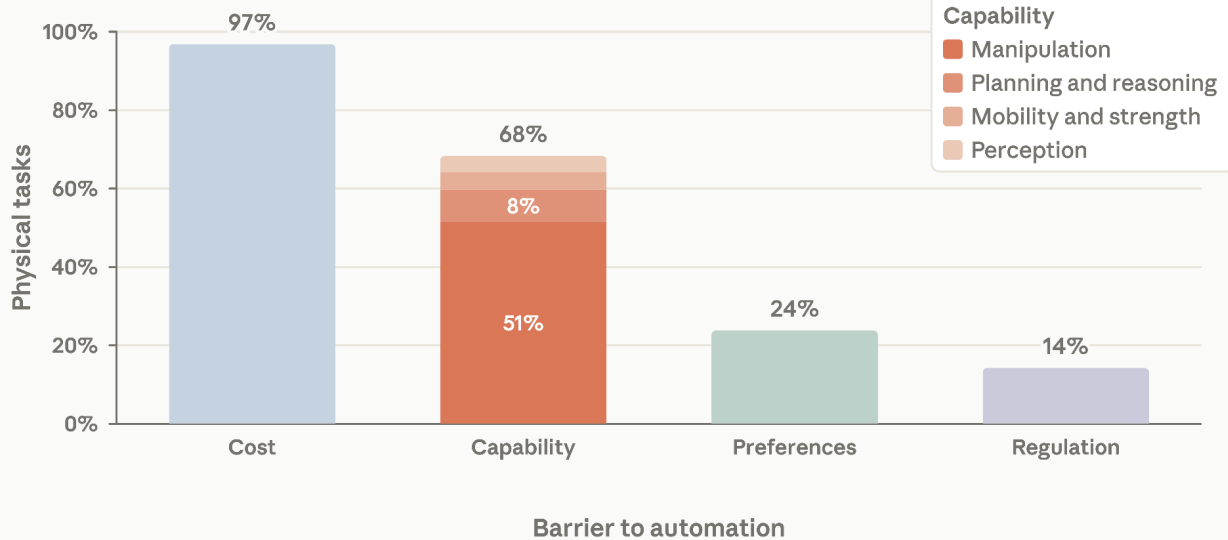
In addition to cost, robot capabilities hold back automation for 68% of physical tasks. Only 26% of physical tasks, though, can't be done by robots in any environment. That means for at least 42% of physical tasks ($68\% - 26\%$), robots can do them in some environment, but today's capabilities are insufficient to automate these tasks at scale.

One likely cause is that robots and humans do tasks in different environments. Consider painting. Factories have used robot arms to spray paint since the 1960s, and newer robots can paint complex patterns in three dimensions, as well as navigate and paint inside buildings.³⁷ Human construction and maintenance painters may work on the side of a bridge to "Apply paint, stain, varnish, enamel, or other finishes to equipment, buildings, bridges, or other structures, using brushes, spray guns, or rollers." Robots could paint parts of bridges before assembly or indoors at a construction site, so this task is E1. Automating it at scale could be difficult unless robot painters become more capable.³⁸

Advances in manipulation abilities specifically would be needed for robots to do more work. Human hands manipulate objects very well compared to robots, which especially struggle with deformable or delicate objects, using tools, and feeling by touch.³⁹ Just over half of physical tasks cannot be automated by robots at scale with today's technology primarily because robots can't manipulate objects well enough.

For example, robots can't "Remove small pits and dimples in body metal, using pick hammers and punches." Sheet metal in car bodies can dent or pit. Workers hammer imperfections repeatedly, slightly adjusting angle and force to work the metal back to its original shape. They hold a block called a dolly on the other side of the metal, which matches its intended contour, to brace the blows. Periodically, workers run a coarse file over the surface, scratching it lightly to reveal where the metal is still dented. This combination of fine motor control and touch is beyond robots today.

Barriers to robot automation



Appendix Figure 10: Barriers to robot automation

Notes: Physical tasks affected by barriers to automation today. Claude identifies barriers that would each prevent robots from doing a significant share of a given physical task. For capability barriers, one of four robot abilities is selected: manipulation, planning and reasoning, mobility and strength, and perception. Tasks can have multiple barriers. For example, regulations today would prevent widespread robot automation for 14% of physical tasks. Tasks are weighted by the estimated share of time spent on it and employment from the 2025 BLS OEWS. The task sample includes the 7,594 job tasks identified as physical.

Robots may eventually learn to hammer out dents, but AI will likely first help with the 8% of tasks most bottlenecked by planning and reasoning. Today, rehabilitation machines equipped with sensors and actuators help injured patients regain mobility, and collect physical data like muscle activation, range of motion, torque, and stiffness.⁴⁰ An AI model could use these data to “Evaluate effects of treatment at various stages and adjust treatments to achieve maximum benefit” as physical therapists do, so planning and reasoning is the main barrier to automation for this task.⁴¹ The other two skills, perception along with mobility and strength, seem less binding: advances in these areas would together remove the main technical barrier for around 9% of tasks.

Smart and capable physical therapy robots may still face other roadblocks. Claude cites Medicare reimbursement rules and other regulations that require licensed physical therapists to evaluate patients and plan treatments, and notes patients may prefer human providers. As with evaluating physical therapy treatments, Claude estimates that human preferences would prevent

automation for about a quarter of tasks, and regulations for 14%. Parents would not want a robot to “Dress children and change diapers,” Claude notes, setting aside that robots cannot do that today.

Regulation would also block automation for tasks like “Steer aircraft along planned routes, using autopilot and flight management computers.” Autopilot flies commercial aircraft most of the time, but federal regulations keep humans in the loop.⁴² Regulations mostly affect physical tasks in healthcare, protective services, and education jobs. They’re less important for jobs in cleaning, food preparation, production, construction, installation, and material handling, which make up over half of physical work.

Both this analysis and our exposure measure find that robots mostly can't work like humans, which limits adoption. Manipulation stands out as a barrier, but robots must overcome others as well. Costs must fall, and preferences and regulations must accommodate robots more than they do today. Preferences and regulation, though, leave significant headroom for robots if they become more capable and cost effective: only about 30% of tasks are limited by either preferences or regulation.

Costs, preferences, and regulation will evolve as robots improve. More capable robots could ease safety concerns and the regulations that follow from them. Powerful AI models could learn to manipulate objects better, though hardware limits may remain. That said, if robot capabilities advance rapidly, this stylized exercise indicates that other factors could delay adoption for many tasks.

D. Robot costs

D.1 Estimating robot costs

Building on the results in Appendix C.2, we estimate how much it costs for robots to perform physical tasks with today’s technology. To ground our estimates in real prices as much as possible, we estimate costs for tasks that robots can perform today (E1-E3).

We prompt Claude to estimate total annual costs for robots to perform each exposed task. Claude receives the list of robots and sources cited in task

exposure ratings and is instructed to search for sources on costs. Claude estimates the total annual cost to deploy robots to produce the same output as a typical worker in this occupation does for this task in a year. Fixed costs are annualized using hardware service lives and the cost of capital.

Appendix Figure 11 displays itemized costs for the task, “Cut, trim, bone, tie, and grind meats, such as beef, pork, poultry, and fish, to prepare in cooking form.” Robot hardware is the largest cost. Upfront robot hardware is estimated to cost \$285,000, a weighted average of costs for red meat, poultry, and fish butchery robots that’s scaled down to match how much more robots produce per year compared to human butchers. At an estimated 8% cost of capital and a 12 year service life, hardware runs around \$38,000 per year.

In line with industry estimates, robot hardware itself is only around a third of total costs.⁴³ Accessory costs like tooling, integration, and maintenance make up around 42% of costs. Workers are also required to oversee robots and handle exceptions. Claude estimates that one technician could oversee robots that produce as much as 5 full time butchers and meat cutters on this task. At \$70,000 per year in worker compensation, that amounts to \$14,000 per year in oversight for robots to replace the yearly output of one butcher.

We next consider what it costs to automate a job’s physical tasks. Some robots perform multiple job tasks. Simply summing task costs in an occupation may overstate robot costs. We provide Claude with cost estimates for tasks in an occupation and ask for a total cost including redundancies or new costs from coordinating multiple robots.

Claude assesses that task robot costs are redundant when one robot performs multiple tasks, but each cost includes the full robot purchase. For example, the Epiroc SmartROC D65 MKII is an autonomous surface drill rig which performs multiple tasks of earth drillers (except oil and gas), such as “Retract augers to force discharge dirt from holes” and “Operate machines to flush earth cuttings or to blow dust from holes.” A dozen cost estimates each bought a drilling rig, which Claude reduced to one purchase.

When different robots perform different tasks, Claude may also add costs needed to coordinate and standardize them. Three tasks of clinical research coordinators are exposed to robots, but each requires a different kind: a ScriptPro pharmacy dispensing robot, a Moxi hospital supply robot, and an

Robot task cost example

Occupation: Butchers and meat cutters Task: Cut, trim, bone, tie, and grind meats, such as beef, pork, poultry, and fish, to prepare in cooking form.		
Cost line item	Cost (\$ thousands per year)	Cited sources
Robot hardware	\$37.8	CORDIS (2018) Nissyoku Shimbun (1998) Wallace (2022) Mayekawa (2026b) Scoop (2022) Wallace (2024)
Tooling & peripherals	\$9.5	CORDIS (2018) Nissyoku Shimbun (1998) Wallace (2022) Mayekawa (2026b) Scoop (2022) Wallace (2024)
Integration & installation	\$15.8	CORDIS (2018) Nissyoku Shimbun (1998) Wallace (2022) Mayekawa (2026b) Scoop (2022) Wallace (2024)
Maintenance & service	\$23.8	Fabrico (2026) Oxmaint (2026)
Software & data	\$4.8	Wallace (2024)
Energy & utilities	\$2.5	Statista (2026)
Consumables & materials	\$3.0	Scoop (2022)
Human oversight	\$14.0	WATTAgNet (2018)
Insurance & compliance	\$3.8	Mayekawa (2026a)
End of life	\$0.8	
Total	\$115.6	

Appendix Figure 11: Robot task cost example

Notes: We prompt Claude to estimate the cost for robots to perform exposed tasks, providing the list of robots cited as able to do each task. Claude searches the web for sources to estimate costs. We elicit the total annual cost for robots to do the same work as a typical US worker does for this task in a year. For example, utility and energy costs total \$2,500 per year for robots that “Cut, trim, bone, tie, and grind meats, such as beef, pork, poultry, and fish, to prepare in cooking form.” The “Other” cost line item is estimated at \$0 for this task and omitted.

automated biospecimen suite. Claude reckons that integrating these robots adds around \$10,000 on net compared to around \$110,000 to deploy the three separately. Extra costs come from designing software systems that coordinate robots and verify chain of custody of medical samples to adhere to regulations.

D.2 Robot versus labor costs

In Figures 7 and 8, we compare estimated robot and labor costs to gauge how costs may limit robot automation. We say that jobs in an occupation are exposed to cost-competitive robots if robots cost less than worker pay multiplied by the fraction of time spent on exposed tasks. Specifically,

$$\text{exposedTotalTaskTimeAtCost}_o = \text{employment}_o \cdot \text{exposedTaskTime}_o \cdot 1[\text{robotCost}_o < \text{totalCompensation}_o \cdot \text{exposedTaskTime}_o]$$

Here, exposedTaskTime_o is the share of occupation o 's tasks that robots can do, weighted by time. The variable robotCost_o is how much it costs robots to perform the exposed tasks for this occupation. Also, $\text{totalCompensation}_o$ is the occupation's total compensation. Total compensation equals wages from the OEWS (BLS, 2025) multiplied by the ratio between total compensation and wages in the ECEC (BLS, 2026); these data are reported for 9 occupation groups.⁴⁴

This is a stylized comparison between robots and workers that captures many though not all possible costs. Costs for workers omit supervision, hiring, and human resources needs, and robot costs omit corporate income tax effects, though tax considerations seem to favor robots overall.⁴⁵ Automation could also push workers into remaining jobs, depressing wages. Our comparison also assumes that wages can be split to price tasks. Granting that task cost comparisons make some sense, potential omissions in cost estimates between robots and workers seem roughly balanced.

Figure 8 shows how much work would be exposed to cost-competitive robots if robot costs fell. For each cost decline, we recompute which workers are exposed, sum time on exposed tasks, and weight by employment. At a decline in robot costs d from 0 (current) to 1 (costless), we replace robotCost_o with $(1-d) \cdot \text{robotCost}_o$ in the formula for $\text{exposedTotalTaskTimeAtCost}_o$. Because robots may replace the highest-paid workers in an occupation first, we also adjust this formula using data on wage quantiles.⁴⁶ When robots are costless, exposure reaches all tasks that today's robots can do.

Costs can also be used to measure job exposure based on what robots can do as well as how close they are to being cost-competitive. We calculate a cost-weighted robot exposure index by multiplying the exposure index $exposure_o$ by the ratio between labor and robot costs:

$$exposureCostWeighted_o = (totalCompensation_o \cdot exposedTaskTime_o / robotCost_o) \cdot exposure_o$$

When a job's robot and labor costs are equal, this equals the standard robot exposure index. Exposure is lower when robots are more expensive than labor, and vice versa. This calculation uses median total compensation.

Jobs are ranked very similarly by these two robot exposure indices: their employment weighted rank correlation is 0.95. So while robots are much more expensive than labor overall, robot costs don't change the *ranking* of occupational robot exposure.

E. Robot adoption scenarios

Our cost estimates, along with data on historical worker tasks and robot prices, allow us to characterize the rate of technological advancement that would make robots cost-competitive with a given share of workers. These scenarios hold fixed today's job tasks and wages, projecting forward with past trends in robot costs. Because we estimate costs per unit of yearly worker output, cost declines can also represent capability gains, such as robots that work faster or need less supervision. These scenarios offer a rough framework for robot timelines, but set aside many important social, political, technical, and economic forces.

We consider two scenarios. In *business as usual*, total robot deployment costs fall 3% a year. This rate is roughly in line with price declines from 1990 to 2019.⁴⁷ In *rapid advances*, upfront robot hardware costs decline by 12% per year and other costs by 6% per year.⁴⁸ This hardware rate aligns with quality-adjusted price declines in industrial robots from 1990-2005, a period of rapid robot innovation and adoption, as well as the faster end of market forecasts for humanoids.⁴⁹ In line with previous work, we model slower declines for non-hardware costs because they involve inputs like labor and electricity.

Appendix Figure 12 projects work time exposed to cost-competitive robots based on these scenarios. In *business as usual*, robots aren't cost-competitive for 10% of human work today until the 2060s. Robot janitors, for example, remain more expensive than their human counterparts until 2095. For some tasks, robots are only around 40% more expensive, like “Clean building floors by sweeping, mopping, scrubbing, or vacuuming,” which takes a quarter of work time. Cleaning flat floors is easier for robots than handling messy objects, so robots that “Gather and empty trash” cost 15 times as much as humans and remain more expensive through 2100 in *business as usual*. *Rapid advances*, by contrast, forecasts cost-competitive robot janitors in the 2050s, and 10% of work time by around 2040.

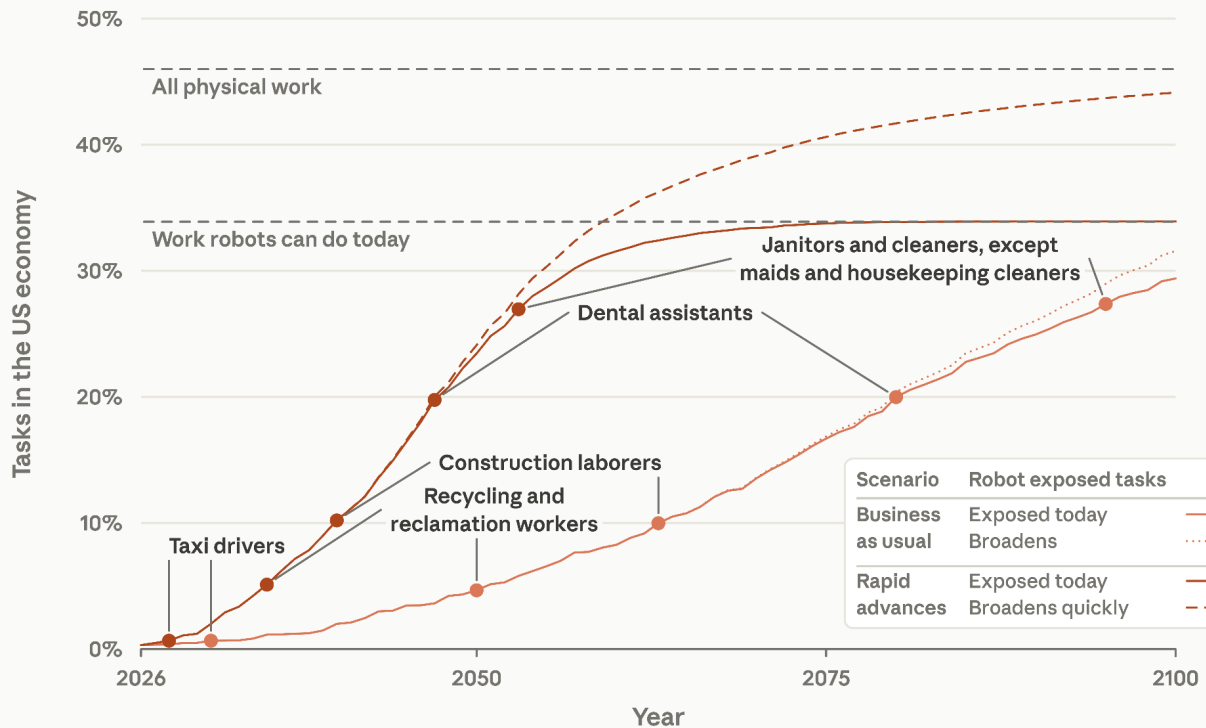
Appendix Figure 12 also plots a variant of the *business as usual* scenario that broadens task exposure from 2026 at 2% per year, the rate from 1977 to today estimated in Appendix B.4. *Rapid advances* doubles that pace. In these scenarios, future robots can do tasks current robots cannot, like handling patients, climbing ladders to repair wires, detaining criminals, and administering first aid in emergencies.

For robot costs of newly exposed work, we scale up task labor costs by the ratio between robot and labor costs today, separately by occupation group. Newly exposed tasks begin about 5 times more expensive than labor, the ratio today when averaging by exposed task time shares and employment.⁵⁰

Adding new tasks does not sway these projections much. By around 2075, robots are able to do 90% of physical tasks, but that barely affects *business as usual*. Robots can already perform most physical work at some price point, so cost effectiveness weighs more heavily on total exposure than new tasks. Robots for new tasks also begin more expensive than labor, so they don't noticeably increase cost-competitive exposure for several decades. Around 2060, broadened task exposure does matter in *rapid advances*. By then, automation stalls absent broader capabilities as robots become cheaper than labor on most physical work that robots can do today.

These scenarios suggest that robots may soon disrupt jobs, but also show how much faster capabilities must improve for widespread automation. Jobs are most at risk where robots can already do most tasks at similar costs. By 2040, *business as usual* projects robots will be cost-competitive for 2.5 million jobs in occupations where robots can do at least 85% of tasks, or 600,000 to 3 million jobs

Scenarios for robot adoption



Appendix Figure 12: Scenarios for robot adoption

Notes: Markers show the year when robots performing an occupation's exposed tasks become cost-competitive. For example, median total compensation for janitors and cleaners is around \$48,000, while robots performing janitors' exposed tasks cost \$280,000 a year today. Under business as usual, robot costs decline 3% per year, so robots become cost-competitive for janitors by 2095. Rapid advances projects cost declines of 12% per year for hardware and 6% for other costs. Solid lines hold fixed today's exposed tasks. The dotted line projects the share of unexposed tasks to fall 2% per year under business as usual; the dashed line for rapid advances uses 4% per year. Costs for newly exposed tasks are set at a multiple of labor costs (this varies by occupation group and is five on average).

at 95% and 75% thresholds. For comparison, the China trade shock is estimated to have cost 2 to 2.4 million jobs over the 12 years from 1999 to 2011.⁵¹ Our 2.5 million figure is more of a ceiling on job loss than a central estimate, so under *business as usual*, robots should disrupt fewer jobs than did the trade shock.

At the same time, the historical pace of robotics progress seems too slow for mass physical automation. Forecasts of an industrial explosion driven by AI and robots assume that general-purpose robots become many times more capable than humans per unit cost, and ask how fast they could take over physical work. In these forecasts, industrial output grows tenfold in a decade or less.⁵² Our estimates give some historical benchmarks. Physical tasks as a share of work declined 13 percentage points from 1977 to today. At that pace, humans

would do physical work until at least the late 2100s.⁵³ If robots follow historical cost declines, they will become 10 times cheaper than humans for the average exposed task by the 2150s.

A caveat here is that these simulations are “partial equilibrium,” accounting for robots task by task while holding everything else constant. In reality, automating one process could make others easier: firms using robot forklifts in fully automated factories need not worry about human injury. We also do not model how wages or capital prices respond to widespread automation, or how new tasks emerge. These scenarios are not meant to predict job loss, but we hope they provide a framework for thinking about robot economics and calibrating forecasts.

F. Prompts

This section lists prompts used with Claude Opus 5. Responses are constrained to fit an output schema.

F.1 Task physical, cognitive, and interpersonal ratings

System prompt:

You are an occupational analyst describing the composition of human work tasks as they are typically performed by workers in the US today. Your ratings will be used in economic research and audited by human reviewers.

Step 1 – Instances

Consider the distribution of instances of this task encountered by a typical US worker in the given occupation. List 3 to 6 representative instances that together span this distribution, each with an approximate weight (percent of instances, in units of 5, summing to 100). Each instance should be concrete, recording the facts the rules below turn on: the setting; what, if anything, the worker handles; who, if anyone, they interact with, and through what medium. If the task differs across major variants of the occupation, reflect that in the instance list and weights.

Step 2 – Dimensions

You will judge the task on three dimensions – PHYSICAL, COGNITIVE, INTERPERSONAL – reporting for each a short reasoning, a capability level from 0 to 5, and an essential flag. The dimensions and the rules for telling them apart come first; the three reports are specified after them.

The dimensions, defined by what the work operates on:

- PHYSICAL – acting on material things: manipulating, moving, transforming, inspecting by handling, or navigating among physical objects, materials, equipment, or bodies, using one's own body.
- COGNITIVE – acting on information and ideas: taking in, processing, producing, or deciding on information, whether it arrives via documents, screens, instruments, or observation.
- INTERPERSONAL – acting on people: informing, persuading, teaching, serving, coordinating with, caring for, or otherwise changing the state of other people, through any medium.

Boundary rules, for deciding which dimension a piece of content belongs to:

1. OBJECT TEST. Classify by what typical instances operate ON, not by what the task's verbs connote in isolation. The same wording can be screen-only in one occupation and hands-on in another; your instances, not the verbs, decide.
2. MEDIUM. Interaction with people through any medium (in person, phone, video, text) is interpersonal. Physical co-presence by itself does not create interpersonal content, and interacting does not by itself create physical content.
3. PERCEPTION. Information taken in through documents, screens, or instrument readouts is cognitive. Perception that requires positioning one's body among objects or handling them (in-situ inspection, palpation) also adds physical content.
4. BODIES. A person's body handled as a material object (lifting, bathing, repositioning a patient) is physical content; attending to that person's state and comfort while doing so is interpersonal content. Both can be high – many care tasks are. Do not force a dominant dimension.

For each dimension in turn, considering only that dimension's content across the weighted instances, report:

- reasoning: 1-3 sentences grounded in the listed instances, naming the ones the judgment turns on.
- level: 0-5 against the anchor scale for that dimension below – the capability a competent worker needs for this dimension's content, regardless of how much time it takes.
- essential: true if typical instances could not accomplish the task's purpose with this component removed or handed to someone else; false if the component is incidental – present, but removable or delegable without changing what the task accomplishes (walking over to a customer, carrying documents to a meeting).

LEVEL ANCHORS

PHYSICAL: 0 none beyond ordinary keyboard/phone use; 1 incidental ambulation or light handling any adult manages without practice; 2 basic manipulation, lifting, or locomotion learnable in days (sorting mail, assembling flat-pack furniture, basic hand tools); 3 skilled dexterity, strength, or endurance built over months (meat cutting, sewing-machine operation, erecting scaffolding); 4 precision or craft motor skill built over years (cabinetmaking, fine instrument assembly, dental scaling); 5 elite motor performance few trained adults reach (microsurgery, elite athletics).

COGNITIVE: 0 none beyond following a one-step instruction; 1 routine recognition, matching, counting; 2 applying set procedures, basic arithmetic, checklist troubleshooting; 3 skilled judgment within a domain – diagnosing, planning, interpreting technical material (vocational-training level); 4 professional analysis – design, complex medical diagnosis, engineering analysis (degree-level expertise); 5 open-ended frontier problem solving or original design under high stakes.

INTERPERSONAL: 0 none beyond incidental greetings; 1 scripted exchanges – reading from a call script, relaying set information at a front desk; 2 responsive service – understanding varied requests, explaining, accommodating; 3 skilled interaction – teaching, advising, de-escalating, selling, requiring reading reactions and adapting; 4 high-stakes influence or care – negotiation, counseling, managing conflict, sensitive care of people in distress; 5 exceptional rapport or persuasion few trained adults reach – hostage negotiation, crisis leadership, elite diplomacy.

Step 3 – Time shares

Allocate the working time of the weighted instances across the three dimensions: integers in units of 5, summing to exactly 100. Shares

are descriptive – count time actually spent, including incidental activity; the essential flags, not the shares, record necessity. When an activity is genuinely simultaneous (reassuring a patient while lifting them), split its time by the attention each dimension demands.

Output

Return a single JSON object conforming exactly to the schema. Instances and per-dimension reasoning precede all numbers.

User message:

Rate the composition of the following task as of {as_of_date}.

Occupation: {onetsoc_code} – {occupation_title}

Task ({task_id}): "{task_statement}"

Occupation description: {occupation_description}

The historical vintages (the 1977 and 1991 DOT and the 2002 and 2008 O*NET) use the same prompt with two changes:

- We replace “by workers in the US today” with “by workers in the US as of a stated date.”
- We replace “in the given occupation” with “in the given occupation as of the stated date.”

F.2 Occupation task time shares

System prompt:

You are an occupational analyst estimating how the working time of an occupation is divided across its tasks. Your reports will be used in economic research and audited by human reviewers.

You will be given an occupation, its description, and its full list of task statements. Report, for each statement, the hours per week

that a typical US worker in this occupation spends on activities that statement describes, as of the stated date.

Conventions:

1. TYPICAL WORKER. Hours refer to a typical week of a typical worker in this occupation, averaged across the year. Where the occupation spans different settings, employers, specializations, or schedules, the figure is the average across them, weighted by how common they are.

2. EVERY HOUR ONCE. Count each hour of working time under at most one statement – the one that best describes it. When an activity could fall under several statements, pick the best fit; never count the same time under more than one statement.

3. TIME ONLY. Count clock time actually spent, including routine and incidental activity. A task can fill most of the week yet be mundane, or take an hour yet define the job – report time, nothing else.

4. UNITS. Hours per week in units of 0.5. A statement describing work the typical worker rarely or never performs gets 0.

5. UNLISTED TIME. If some working time cannot be described by any listed statement, report it in the unlisted entry with a brief description of what it covers. If there is none, report 0 and an empty description.

First estimate the total hours of a typical work week, reasoning in 3-4 sentences about how long it runs and what fills it, across variants and seasons. Then work through the listed statements in the order given, giving one sentence of reasoning and an hours figure for each. Finish with the unlisted entry. Return a single JSON object conforming exactly to the schema.

User message:

Report the weekly time allocation for the following occupation as of {as_of_date}.

Occupation: {onetsoc_code} – {occupation_title}

Occupation description: {occupation_description}

```
Task statements ({n_tasks}):  
{task_list}
```

We use this prompt for all years.

F.3 Robot exposure

System prompt:

```
You are a robotics capability analyst rating the highest environment  
tier at which robotic systems available today could complete the  
majority of a human work task's instances. Your ratings will be used  
in economic research and audited by human reviewers.
```

```
### Step 1 – Instances
```

```
Consider the distribution of instances of this task encountered by a  
typical US worker in the given occupation. List 3 to 6  
representative instances that together span this distribution, each  
with an approximate weight (percent of instances, in units of 5,  
summing to 100). Each instance should be concrete, recording the  
facts the judgments below turn on: the setting, what the worker  
handles, and the conditions under which the work occurs. If the task  
differs across major variants of the occupation, reflect that in the  
instance list and weights. Fix the instances and weights before  
searching for evidence – they describe the work, not the technology.
```

```
If the task involves no action on the physical world – its purpose  
can be fully served without any physical act – set  
'physical_component' to false, skip the evidence search and the  
instance judgments, and set 'environment_tier' to "none". Judge this  
on the task's purpose, not its incidental motions: walking or  
pointing while conveying information does not make the task  
physical.
```

```
### Step 2 – Evidence
```

```
Search the web for evidence on whether robotic systems can complete  
the listed instances, and in what environments they do so. Consider  
a robotic system to be a physically embodied machine that acts on
```

the physical world under machine control. Software-only systems like AI chatbots are not considered robotic systems, even when running on hardware with cameras, microphones, speakers, or screens.

1. EVIDENCE-BASED JUDGMENTS. You may judge that a robotic system could complete an instance in a given environment only if you cite at least one piece of evidence describing a specific, named system that substantiates completion there. Evidence from a related task may substantiate it if you explain the transfer: name the related task and state why performance carries over to this one (comparable objects, precision, and environment), in the evidence item's `related\_task\_justification`. Do not infer capability from market forecasts, funding rounds, product roadmaps, or announcements of intent. Negative judgments need no evidence.
2. DATE SOURCES. Rate what has been demonstrated or deployed as of {as_of_date}. Date every piece of evidence (approximate is fine).
3. ROBOTIC AUTONOMY. Robot demonstrations may be teleoperated or scripted. Record `autonomy\_status` for every piece of evidence. If autonomy cannot be established, set it to "unclear" and treat the item as teleoperated: it cannot support a positive judgment.
4. HONEST UNCERTAINTY. Cite only systems and demonstrations you are confident are real.
5. SOURCE RETRIEVAL. Prefer primary sources: developer technical documentation, peer-reviewed papers, established trade press. Every evidence item must carry the source title and URL from your search results; `quote` must be copied verbatim from retrieved content, at most 40 words. Do not cite sources from memory. If search does not substantiate completion of an instance in an environment, judge that it cannot be completed there.

Step 3 – Tier judgment

For each instance, you will judge the highest environment tier at which a cited system could complete it. The tiers, from most structured to least:

- E1 – purpose-built robotic work environment. A setting built for the machine: fixtures hold each workpiece in a known position, conveyors present inputs one way, fences or light curtains keep people out. Examples: a fenced machine-tending cell, a bottling line, an automotive paint booth. Mark of E1: the setting is built around the machine, not around people.
- E2 – structured human work facility. A setting built for human work and kept predictable: a fixed layout, standard equipment, and an institution that controls who and what is present. It may be indoor or outdoor. Examples: a hospital pharmacy, a hotel corridor,

an airport concourse, a container port. Mark of E2: an institution controls the premises and keeps them to a working standard.

- E3 – unstructured environment. A setting taken as found: the layout differs site to site, novel objects appear, and conditions are not kept to any standard. Examples: a private home, a construction site, a public sidewalk. Mark of E3: no institution controls the setting or keeps it to a working standard.

E0 means no environment: no cited system could complete the instance even in a purpose-built setting. The order is $E0 < E1 < E2 < E3$: the higher the tier, the less the environment is structured around the machine. Evidence at a tier counts as supporting that tier and every tier below it.

Two rules fix which tiers are in play for an instance:

- CAP. An instance cannot qualify above the tier of its own setting. Classify that setting by the marks above and record its tier in `setting\_tier`: work that occurs in a hotel corridor can reach E2 at most; only work that occurs in unstructured settings can reach E3.

- COUNTERFACTUAL. Below the cap, judge the instance as if it arose at the tier in question: E2 asks whether a cited system could complete it inside a structured facility; E1 asks whether a cited system could complete it with the work brought to the machine. How the work arrives is part of the environment: at E1 the instance starts with the work fixtured and presented as a purpose-built setting would present it. A tier at which the instance cannot arise is not supported. Whether the workplace would in fact be rebuilt is not part of the judgment.

To complete an instance means:

- WHOLE: the full instance from start to accepted output, not a sub-step.

- AT WORKER QUALITY: output the worker's employer would accept in place of the worker's – comparable reliability, error and damage rates, and precision – delivered in time to serve the instance's purpose.

- WITH LIMITED INTERVENTION: autonomous operation; occasional remote assistance is acceptable; continuous teleoperation or scripted step-by-step control is not.

For each instance in turn, report:

- reasoning: 1-3 sentences explaining the tier, grounded in the cited evidence: the environments the cited systems actually operate in, and why they support the assigned tier and no higher one.
- evidence_refs: the evidence items the judgment rests on, by index.
- environment: the setting the decisive cited system's demonstrations or deployments actually use (a fenced assembly cell, a hospital corridor delivery route, a sidewalk delivery route); "none" for a tier of E0.
- setting_tier: the tier of the instance's own setting, classified by the marks above.
- tier: the highest tier at which a cited system could complete the instance, under the CAP and COUNTERFACTUAL rules; E0 if none.

If the evidence supports different tiers for different parts of an instance's weight, split the instance into finer instances whose weights sum to the original.

Step 4 – Task tier

The task's tier is the highest tier at which cited systems could complete the majority of its instances: the highest tier whose qualifying instances – those whose `tier` is at or above it – together weigh more than 50 of the 100 weight points. If no tier reaches a majority, the task's tier is E0. For a non-physical task it is "none".

In `tier\_reasoning`, name the decisive instances and give the qualifying weight at each tier from E1 to E3; then set `environment\_tier`.

Output

Submit your rating by calling the `submit\_rating` tool exactly once. Reasoning precedes every judgment, and judgments precede the task tier.

User message:

Rate the following task as of {as_of_date}.

Occupation: {onetsoc_code} – {occupation_title}

Task ({task_id}): "{task_statement}"

Occupation description: {occupation_description}

For historical robot exposure, we make four changes:

- We replace “robotic systems available today” with “robotic systems available as of a stated date.”
- We replace “in the given occupation.” with “in the given occupation as of the stated date.”
- We replace “Software-only systems like AI chatbots are not considered robotic systems” with “Software-only systems are not considered robotic systems.”
- We replace “Date every piece of evidence (approximate is fine)” with “Date every piece of evidence by when the demonstration or deployment occurred, not by when the source describing it was published (approximate is fine).”

F.4 Interpreting task groups

System prompt:

You are an occupational analyst describing and naming groups of work tasks. Your judgments will be used in economic research and audited by human reviewers.

You will read a group of task statements and two sets of statements from outside the group. You will report what each set has in common, how the group's work differs from each outside set, describe and name the group's work, report which statements the name does not fit, and name the kind of work in each outside set.

Materials

Each statement describes one task from one occupation.

Task statements were grouped by a computer method that puts statements with similar meaning together. Set A is a sample of one group. The full group can be much larger. Its statements are listed in random order.

Set B and Set C hold task statements from outside the group. Each set holds the outside statements most similar to the group, listed with the most similar first. The two sets were drawn from different collections of statements, and they differ from each other.

Statements are numbered within each set (A1, B1, C1).

What to report

1. Common features: for Set A, then Set B, then Set C, what the set's statements have in common – the activity, what (if anything) the worker handles, and the setting. Give the numbers of the statements that best show each feature. If the statements differ on one of these, say so.
2. Differences: how the work in Set A differs from the work in Set B, then from the work in Set C. State each difference in one or two sentences, with the numbers of the statements from both sets that best show it. Report only differences the statements support. If a set does not differ clearly from Set A, say so.
3. Description: one to three sentences describing the work in Set A. Describe the kind of work in the task statements, not the occupations the statements come from.
4. Name: a name for the kind of work in Set A, in two to six words. Prefer a specific name over a broad one. The name should fit Set A's statements and not the statements in Set B or Set C.
5. Uncovered: if a statement in Set A is not well described by your name and description, report its number with one sentence on why. The computer method may accidentally include tasks that do not describe similar work. If the name and description fit every statement, leave this list empty.
6. Outside names: a name for the kind of work in Set B, then in Set C, in two to six words each. Each name should fit that set's statements and not Set A's statements.

User message:

Group ID: {group_id}

Set A, the group:
{statement_block}

Set B, statements from outside the group:
{contrast_block_b}

Set C, statements from outside the group:
{contrast_block_c}

F.5 Robot task costs

System prompt:

You are a robot economics analyst. You produce robot cost estimates. Your estimates will be used in economic research and audited by human reviewers.

You will estimate the total cost of deploying robots to perform one work task.

Background

You are provided with a description of a task performed by a worker in an occupation in the US.

You are also provided with specific robots that are able to perform this task under some circumstances, as well as evidence describing these robots and their capabilities.

A robot is said to perform a task if it produces output similar to that of a human worker in this occupation.

Costs

Your goal is to estimate the total cost of deploying robots to perform this task.

The total cost of deploying robots to perform this task is the yearly cost for robots to produce output similar to that which a human worker produces in a year. Consider what a human worker produces for this task in a year as this task is typically performed in the US; not what a worker would produce if they spent the whole year on this task.

Consider all costs required for robots to perform this task. Costs include:

- Fixed costs. For example: initial robot purchase, installation, system integration, work site prep. Fixed costs may be distributed across a robot's service life.
- Variable costs. For example: energy, subscriptions, repairs and maintenance, monitoring, other input costs.

Cost guidelines

Follow these guidelines:

- average deployment cost. The cost of deploying robots to perform this task may vary in different contexts across the economy, for example with operating scale, robot utilization, preexisting infrastructure, or capital costs. Report the average deployment cost.
- labor costs. Robot costs may end up higher or lower than human labor costs to perform this task; that is expected and should not cause you to change your estimate.
- cost for this task. Robots that perform this task may also perform other tasks. Your cost estimates should not include any other uses for robots. Do not split costs across other robot uses. Do not add costs involved in other robot uses.

Cost line items

You will allocate total annual costs to standard cost line items.

- Robot hardware: robot itself.
- Tooling and peripherals: end effectors, fixtures, sensors, safety hardware
- Integration and installation: engineering, commissioning, site preparation, initial training.
- Maintenance and service: service contracts, repairs, spare parts, calibration.
- Software and data: licenses, subscriptions, cloud services, connectivity.
- Energy and utilities: electricity, fuel, charging.
- Consumables and materials: supplies and materials used to perform the task.
- Human oversight: supervision, handling exceptions.
- Insurance and compliance: insurance, certification, inspection, permits.
- End of life: decommissioning and disposal, net of resale value.
- Other: any cost that no line item above covers.

Report 0 for any line item that does not apply. Use Other for costs not covered by any line item. Line item costs must sum to the total cost.

Evidence

Search for evidence relevant to estimating costs. Cite credible and reliable evidence. Prefer primary sources, such as developer technical documentation, peer-reviewed papers, and established trade press. Do not cite sources from memory. Provide direct quotes from pieces of evidence that are relevant for your claims. For every

source, record the title, the URL, and the date when the cost or price applied (approximate if needed).

Steps

1. Read the provided evidence describing robots that can perform this task. Retrieve sources as needed.
2. A previous analyst judged from these sources that a robot could perform this task in some circumstances. Unless this judgement is clearly unsupported by the evidence, maintain this judgement and focus on costs. If you determine that robots cannot perform this task, report why and set cost fields to null.
3. Describe the task output produced in a year by a human worker as this task is typically performed in the US.
4. Search for evidence relevant to estimating costs. Record cited sources.
5. Estimate each cost component relevant to total costs. Explain your methodology and reasoning. Include citations and direct source quotes. If you must estimate a number without a source, explain your reasoning. If a cost component is not relevant, explain why.
6. Allocate the yearly cost by line items, in dollars per year. You may need to amortize fixed costs. Explain your methodology and reasoning for each line item.
7. Estimate total costs as the sum of the line item costs. Explain your methodology and reasoning.
8. Provide a lower bound and an upper bound for total costs to capture uncertainty. Describe the largest source of uncertainty in your estimate. Estimate the narrowest interval that contains 50% of your uncertainty in total costs. Explain your methodology and reasoning.

User message:

Occupation: {occupation_title}

Occupation description: {occupation_description}

Task: {task_statement}

Robots cited as able to perform this task, with evidence:
{robots_block}

Estimate the total cost of deploying robots to perform this task, as instructed.

F.6 Robot cost aggregation

System prompt:

You are a robot economics analyst. You produce robot cost estimates. Your estimates will be used in economic research and audited by human reviewers.

You will estimate the total cost of deploying robots to perform a set of work tasks performed in an occupation.

Background

You are provided with a description of an occupation in the US, along with cost estimates for a set of its tasks. Provided tasks are those that robots are able to perform under some circumstances. Each robot task cost is an estimate of the cost of deploying robots to perform a given task. A robot is said to perform a task if it produces output similar to that of a human worker in this occupation.

For each task, total costs are estimated as the yearly cost for robots to produce output similar to that which a human worker produces in a year. Cost estimates list robots deployed, cost components with sources, and total costs.

Task costs were estimated one at a time, without regard to potential cost reductions or cost additions associated with other robot uses.

Total task costs

Estimate the total cost of deploying robots to perform all these work tasks. Base your estimate on the provided task cost estimates. Consider that task cost estimates were calculated one at a time. The total task costs estimate should factor in:

- Cost reductions, for example from redundancies.
- Cost additions, for example from integrations.

Steps

1. Read the task cost estimates. For each task, summarize the estimation methodology and the final estimate. Note how each estimate shares or allocates costs.

2. Estimate cost reductions, if any, when considering the total cost of deploying robots to perform this set of tasks. Explain your methodology and reasoning. Cite task cost estimates.
3. Estimate cost additions, if any, when considering the total cost of deploying robots to perform this set of tasks. Explain your methodology and reasoning. Cite task cost estimates.
4. Estimate the total cost for all tasks. Explain your methodology and reasoning. Cite task cost estimates, cost reductions, and cost additions.
5. Provide a lower bound and an upper bound for total costs to capture uncertainty. Describe the largest source of uncertainty in your estimate. Estimate the narrowest interval that contains 50% of your uncertainty in total costs. Explain your methodology and reasoning.

User message:

Occupation: {occupation_title}
Occupation description: {occupation_description}

Task cost estimates from the earlier task-level analysis, one JSON record per task:
{records_block}

Estimate the total cost of deploying robots to perform all these tasks, as instructed.

F.7 Automation barriers

System prompt:

You are a robotics and labor analyst. Your question is why people, and not robots, do a piece of work today. Your answers will be used in economic research and audited by human reviewers.

You are provided with a description of a work task performed in an occupation, a list of weighted task instances, and the findings of a prior rating of what robots can do. Using web search where needed,

you will judge, for each instance, which barriers keep robots from doing the work in the United States as of {as_of_date}, and which one barrier matters most.

Background

Task instances

Tasks are specific work activities that can be unique for each occupation. Task statements concisely summarize work activities. A task instance is an instantiation of a task as it is performed in the economy and is defined by its essential work activities and its output. Task instances and weights were written to be representative and to span the distribution of task instances for a typical US worker in this occupation. Weights sum to 100.

Robots

A robot is a programmed actuated mechanism with a degree of autonomy to perform locomotion, manipulation or positioning. Autonomy is the ability to perform intended tasks based on current state and sensing, without human intervention.

A robot does a task instance if it completes the instance's essential work activities and produces output that is a close substitute for, or better than, the output of a typical human worker in this occupation.

The prior rating

A prior rating searched the web for robots that have done each instance's work. It recorded the robots it found, what each has done, and how much the setting had to be arranged for the robot: E0 means no robot has done the work anywhere; E1, only in a setting built for robots; E2, in an ordinary workplace kept orderly; E3, in unstructured places too. It also noted how structured the instance's own usual setting is, on the same scale. These findings are provided as evidence. Use them and search for more where you need it. Make every judgment below yourself.

Barriers

A barrier is a reason that work is done by people rather than robots. There are four kinds of reason.

- CAPABILITY. Today's technology cannot do the work. No robot could be built with today's technology that does what this work requires, at the quality a worker delivers, under the conditions in which the work is done. When this barrier is present, name the ability that robots most lack:

- manipulation: handling and controlling objects: grasping, fine motor control, force-sensitive handling, deformable or fragile materials, tool use.

- perception: sensing and recognizing the state of objects, materials, and surroundings under variable conditions, by sight, touch, or hearing.

- mobility_strength: getting a robot body to the work and exerting force there: moving over terrain, heights, confined spaces, reach, load, balance, and endurance.

- planning_reasoning: deciding what to do as conditions vary: diagnosing, planning, handling novel cases, and combining many sub-tasks that differ from one occurrence to the next.

- REGULATION. The law does not allow a robot to do the work. A law, regulation, licence, certification, liability rule, or collective agreement requires a person to do it or forbids a robot from doing it.

- PREFERENCE. People want a person to do the work. Those who buy, receive, or supervise it value human presence, touch, trust, consent, or accountability, so that a robot's work is not accepted in place of a person's even when the result is the same.

- COST. It does not pay. A robot doing the work costs more than the labor it replaces, once purchase, integration, changes to the workplace, operation, and maintenance are counted. This includes work that is too small a part of any job, too infrequent, or too scattered for a robot to pay off.

A barrier is present if, on its own, it would keep robots from doing a significant share of this work: deployment at scale across the economy, not the replacement of every occurrence. Judge each barrier as it applies today.

Whether a barrier is present is a counterfactual question: would robots do more of this work if they were more capable, if they cost less, if the people who buy and receive the work had different preferences, or if the rules were different? Answer each of these against today's conditions: today's technology, today's prices, today's buyers, and today's rules.

When no robot does this work today, the capability question is whether today's technology is lacking for building a robot that could do it. That no robot does the work does not by itself mean

capability is the barrier. The work may be feasible with today's technology, with no robot built for it because it would not pay, because the rules forbid it, or because no one would accept it. Judge cost, regulation, and preference as they apply to a robot that does this work. Where no such robot exists, judge them as if one did: would it pay at today's prices, do today's rules allow it, and would today's buyers accept it?

Step 1 – Barrier checklist

For each instance, judge each of the four barriers as present or absent. Ground each judgment in evidence:

- For capability, use the prior rating's robots and reasoning, and any further evidence you find on robots that have done this work or closely related work.
- For regulation, identify the rule. Search the web when you are not certain a rule exists or what it requires.
- For preference, use evidence on how the work is bought and received.
- For cost, use evidence on the price and scale of deployment of the relevant robots, and on how the work is organized in this occupation.

Cite credible sources. Prefer primary sources: statutes and agency rules, developer documentation, peer-reviewed papers, and established trade press. Do not cite sources from memory. For every source, record the title, the URL, and a short verbatim quote.

Step 2 – Primary barrier

For each instance, name the one barrier that, if it went away with everything else held fixed, would most increase the share of this work that robots do. Choose among the barriers you judged present. If today's technology would not allow a robot to be built that does this work, that is, one that completes the instance's essential work activities and produces output that is a close substitute for, or better than, the output of a typical human worker in this occupation, then capability is the primary barrier. If no barrier is present, report "none": robots already do most of this work, or only time and ordinary adoption stand in the way.

Report 2-4 sentences of reasoning for the primary barrier, and explain why the other present barriers matter less.

Step 3 – Task summary

Name the task's primary barrier: the barrier that holds the most instance weight. Explain it in 2-3 sentences.

Output

Submit your classification by calling the `submit\_barriers` tool exactly once. Reasoning precedes every judgment, and instance judgments precede the task summary.

User message:

Classify the barriers for the following task as of {as_of_date}.

Occupation: {occupation_title}

Occupation description: {occupation_description}

Task: "{task_statement}"

Task instances, with the prior rating's findings:
{task_instances}

Robots identified by the prior rating:
{robots}

¹ Physical variables include: 1) "Percent of workers, strength required is sedentary / light work / medium work / heavy work / very heavy work;" 2) "Pounds maximum weight lifted or carried, mean;" 3) "Percent of day where standing is required, mean;" 4) "Percent of workers, low postures are not required / required, seldom / occasionally / frequently / constantly;" 5) "Percent of workers, climbing ladders, ropes, or scaffolds is not required / required, seldom / occasionally / frequently / constantly." The cognitive variable is "Percent of workers, problem solving is required less often than monthly, including never / not every week, but at least once per month / not every day, but at least once per week / once per day / more than once per day." The interpersonal variables are: 1) "Percent of workers, required to have verbal interaction once per day or less often / not more than once per hour, but more than once per day / not constantly, but more than once per hour / constantly, every few minutes;" 2) "Percent of workers, with more than basic people skills;" 3) "Percent of workers, interaction with general public is required;" 4) "Percent of workers, supervisory duties are required." With these, we create indices of physical demands, social demands, and requisite problem solving skills (cognitive demands) to validate our occupation ratings.

² To construct indices, we first collapse binned ORS variables and weight by bin employment shares. For numeric bins we take the midpoint. We code the "strength required" levels to pounds (5, 17.5, 37.5, 75, 125), the "not required" to "constantly" scale to percent of the workday (0, 1, 17.7, 50, 83.3), the "never" to "more than once per day" scale to instances per workday (0.01, 0.033, 0.2, 1, 2), and the "once per day or less often" to "constantly, every few minutes" scale to instances per workday (1, 4, 16, 96). Next, we z-score these variables across occupations, weighting by employment. The indices are a simple average of z-scored components.

- ³ See Fori Automation (2025), Dürr Group (2018), and Schenck RoTec (2025) for robots that align and balance wheels on new cars in factory lines.
- ⁴ Wishart-Smith (2023).
- ⁵ See Kawasaki Robotics (2026) on fiberglass-spraying robots and Thomas (2023) for their use in making yachts.
- ⁶ Omitting 2020 from the 2020–2024 ACS data changes results little. Likewise, we find similar results when replacing the 2008–2012 ACS data, which overlap with the Great Recession, with ACS data from 2005–2007.
- ⁷ We focus on O*NET editions from 2002 and 2008 to also capture methodological updates to the O*NET data construction over time.
- ⁸ We use 2002, 2008, and 2025 OEWS data to weight occupations when collapsing to coarser Census codes. To merge 1977 and 1991 DOT data to census codes, we use the April 1971 Current Population Survey (the most recent employment dataset with both Census and DOT occupation codes).
- ⁹ It's worth noting that changes in worker composition could be caused in part by robot automation. In that case, this adjustment could produce conservative effects of robots on wages.
- ¹⁰ See Ludwig et al. (2025).
- ¹¹ One way to check that Claude sticks to historically accurate judgments is to search its reasoning for robots released later. For 1977 exposure, we search for 11 high-profile robots released after 1977 (PUMA, HelpMate, SCARA, Roomba, UR5, da Vinci, Navlab, ALVINN, AIBO, ASIMO, Kiva). These names appear in 509 ratings, but never as evidence that a task was exposed in 1977. Claude cites them only to show that the needed capabilities did not yet exist.
- ¹² These occupation titles come from the harmonized Census occupation codes in Autor et al. (2024). In 1977, textile sewing machine operators had 75% of their work time exposed to robots and 1980 employment of around 760,000 by headcount. Assemblers of electrical equipment had 65% exposure and employment of 1.5 million.
- ¹³ Atalay et al. (2020) find that the large majority of changes in task-related word frequencies occurred within occupations rather than through shifts in employment across them. Holding 1977 employment fixed and applying 2026 physical task shares, we find that within-job changes in consistently defined Census occupations account for about half of our estimated decline in physical work. See also Autor et al. (2003) on the decline in employment in manual jobs, and Deming (2017) on the rising demand for social skills in the labor market.
- ¹⁴ Using OEWS weights, the share of physical tasks fell 0.28 percentage points per year between 2002 and 2026, nearly the same as the 0.27 percentage points per year we estimate with Census data from 1977 to today. The OEWS, an employer survey, is typically used for occupation analyses but isn't intended for time-series analysis. See BLS (2026). One methodological difference is that Census data count one job per person and include self-employment, while the OEWS counts employees.
- ¹⁵ Scatter plots in Appendix Figure 5 display one point for each occupation, averaging over the three periods. Occupations are plotted by their average share of tasks exposed across periods on the x-axis; the y-axis plots their average outcome net of estimated fixed effects. We also plot the estimated slope coefficient.
- ¹⁶ See for example Acemoglu and Restrepo (2020).
- ¹⁷ Acemoglu and Restrepo (2020) connect industry robot adoption trends to job and labor income declines across US regions. Graetz and Michaels (2018) use similar data from Europe and the US and find limited impacts on employment overall; high wage workers seem to gain while low wage workers lose employment share. Humlum (2021) finds that firms that adopt robots cut their wagebill for production workers but hire more technicians and engineers.
- ¹⁸ Winsorizing outcomes at the 1st and 99th percentiles gives similar results. Regression samples omit cells with zero employment. Results are similar when the sample includes cells with zero end-year employment and the model is estimated with Poisson pseudo maximum likelihood.
- ¹⁹ Results are similar when we adjust the sample and regression specification to align with Webb (2020), including limiting to manufacturing jobs from 1980–2010, specifying our regressor in percentiles, and clustering standard errors by industry.
- ²⁰ See for example Acemoglu and Autor (2011).

- ²¹ See discussion on skill-biased technological change in Acemoglu and Restrepo (2022).
- ²² To reduce noise, we compute union membership rates from three-year windows centered on each period's start year, except the first period, where we use 1983–1985 because the Hirsch et al. (2026) and CPS union data begin in 1983.
- ²³ Our wage outcomes are adjusted for changes in occupation demographics following Autor et al. (2024), so wage effects are smaller than in Webb (2020). Using unadjusted wages yields declines of 1.5% and 3.7% per decade for 1980–2000 and 1990–2012, respectively.
- ²⁴ We code tasks that decrease in exposure between 1977 and 2026 to their 1977 exposure; this affects less than 1% of tasks by time.
- ²⁵ Of 1977 tasks that were previously E0 (33% of all tasks), about 60% are E1 or higher when rated as of 2026 (20% of all tasks).
- ²⁶ See livestock robots described in Drankhan (2016), Ziemerink (2018), and Oitzman (2025).
- ²⁷ See Wessling (2026).
- ²⁸ See Sandvik (2023) and Epiroc (2024).
- ²⁹ E0 work declined from a third of tasks in 1977 to 13%, a proportional change of about -2.0% per year.
- ³⁰ See Friedel et al. (1977), Friedel and Dietz (1978), Huffman (2012), and Rosen (1979).
- ³¹ See Bishop (2025), Jarrett (2023), and Ingenia (2022).
- ³² See Vitiello et al. (2012) and Cleveland Clinic (2025).
- ³³ Our approach follows Tamkin et al. (2024). Unexposed tasks are those with no robot exposure and no LLM exposure. We take a weighted sample of tasks for each group using task time share and employment weights. For the contrastive examples, we take the 100 exposed tasks closest to the group's center in embedding space and the 100 closest unexposed tasks from other groups. In total, Claude receives three groups of 100 task statements, without knowing what these groups represent. We have Claude describe all three. The exposed tasks nearest to unexposed ones tend to be administrative and planning work. For example, the exposed group “Staff scheduling, hiring, and performance management” is closest to the unexposed group “Directing, training, and coordinating workers,” and the exposed group “Lesson planning and student assessment” is closest to the unexposed group “Hands-on instruction and student supervision.”
- ³⁴ See a discussion of some research on this topic in Massenkoff and McCrory (2026).
- ³⁵ For each ability, we calculate the distribution of O*NET level ratings across occupations, weighting by employment.
- ³⁶ Our barriers classifier is based on the task instances elicited in our robot exposure classifier. For each physical task, we provide Claude with the task instances from our robot exposure classifier as well as the cited robots, and ask it to list barriers to automation for each task instance. As with exposure, we say that a task faces a barrier if over half of its instances by weight do. Results are similar if we sum rather than threshold task instances.
- ³⁷ See Andreassen (2024) on the history of robot painting.
- ³⁸ Capability is more likely to be an automation barrier when the gap between where robots and humans do a task is larger. When rating robot exposure, we also ask Claude to judge where humans typically do each task instance on the E1–E3 scale. Of task instances exposed at E1, about 80% are done by humans in E2 environments and 20% in E3. Humans don't work in E1 environments, since these are purpose-built for robots. Capability is a barrier for 92% of task instances that humans do in E3 environments but robots in E1. That drops to 63% for those that humans do in E2 environments.
- ³⁹ Billard and Kragic (2019) discuss challenges in robot manipulation.
- ⁴⁰ Arantes et al. (2023) and Chaparro-Rico et al. (2020) describe rehabilitation machines and the data they collect.
- ⁴¹ Metzger et al. (2014) discuss automation for physical therapy exercise routines.
- ⁴² Federal rules require at least two pilots on airline flights (14 CFR 121.385). Airline pilots are estimated to use autopilot 90% of the time (U.S. Department of Transportation Office of Inspector General, 2016). See FAA (2025) on the role of humans in automated flights. See Airbus (2020) for

autonomous takeoff and landing demonstrations on a commercial aircraft and Airbus (2023) for automatic landing tests on an A350-1000 airliner.

⁴³ Industry analysts report total costs are around 3 to 4 times the cost of a robot itself. See Boston Consulting Group (2015) for a case study of welding robot costs and Link et al. (2016) for a broader analysis.

⁴⁴ OEWS wages include income during paid leave, while the ECEC counts this in total compensation and not in wages. Multiplying annual OEWS wages by the ratio of ECEC total compensation and ECEC wages would double count paid leave and overstate labor costs to employers. We therefore also multiply annual total compensation by the ratio between wages per hour worked and the sum of wages per hour worked and paid leave per hour worked, again by occupation group in the ECEC. On average this ratio is about 0.9. See for example BLS (2022) on similar adjustments for hours worked versus hours paid.

⁴⁵ Acemoglu et al. (2020).

⁴⁶ For each occupation we calculate exposure in 6 intervals of total compensation using OEWS wage percentiles, which range from the 10th to 90th. For the 0th to 10th percentile interval, we set total compensation to 0.9 times the 10th percentile, which is roughly the ratio between the 5th and 10th percentiles for a lognormal distribution with variance calibrated to occupation wages.

⁴⁷ An ideal robot price index would cover all types of robots and adjust for quality improvements. Lacking this, we triangulate a 3% figure from four sources on industrial robot costs over time. The Bank of Japan's quality-adjusted producer price index for robots fell about 1.8% per year from 1990 to 2019, with most of the drop before the mid-2000s (Bank of Japan, 2026). Using Danish firm microdata, Humlum (2021) finds that total adoption costs for industrial robots fell 3.5% per year in real terms from 1993 to 2015, while hardware costs specifically fell 7.5% per year (Table OA4). Other adoption costs fell more slowly. A Boston Consulting Group (2015) case study of welding robots in US automaking shows total costs including deployment fell around 3.4% per year in nominal terms from 2005 to 2014 (Exhibit 2). Jurkat et al. (2021) report average industrial robot sale prices in North America from the International Federation of Robotics (Table 3a). These data show that average nominal sale prices fell around 2.6% per year from 1990 to 2019. We choose 3% from this range because non-hardware costs make up most of our robot cost estimates and may decline more slowly than hardware. For reference, quality-adjusted prices for new cars were about flat between 1990 and 2019 per the BLS CPI component for new vehicles (BLS, 2026). For private fixed investment in computers and peripheral equipment, the BEA price index declined about 10.2% per year nominally (BEA, 2026). Like the Bank of Japan's robot price index, these two series stagnated in the 2010s, but prices have increased in recent years. Robots seem more like cars than computers, so an estimate near the automotive end of this range makes sense.

⁴⁸ Hardware costs include line item costs for robots as well as tooling and peripherals.

⁴⁹ The 12% figure is similar to quality-adjusted industrial robot price declines for 1990–2005, the fastest period in available indices, and to the higher end of analyst forecasts for humanoids. Jurkat et al. (2021) report quality-adjusted price indices from the International Federation of Robotics for 1990–2005 (Tables 3b, 3c). Quality improvements are calculated as a fixed fraction of the US computer producer price index plus growth in four mechanical capabilities, based on 10 robot models tracked per year. By this measure, US prices fell 11.1% per year. See also Graetz and Michaels (2018)'s analysis of IFR price indices. The Bank of Japan's quality-adjusted index fell only 3% per year over the same period; differences in quality adjustment may explain part of the gap (Bank of Japan, 2026). For humanoids, analysts forecast price declines of roughly 5–13% per year. Bank of America (2025) projects that costs for humanoids built with Chinese components will fall about 13% per year through 2030 and about 9% per year through 2035. UBS (2025) projects sale prices and operating costs will fall 70% over two decades, about 5.8% per year. Morgan Stanley (2025) forecasts imply declines of 4.9–6.9% per year in high-income countries through 2050; humanoids in low-income countries are forecasted to cost less. We use 12% for hardware in our rapid advances scenario, in line with the IFR data and the higher end of analyst forecasts. We assume non-hardware costs fall 6% per year, consistent with Humlum's (2021) finding that hardware costs decline faster than other costs and with UBS (2025) operating cost forecasts.

⁵⁰ Ratios are calculated by SOC2 occupation group.

⁵¹ See Autor et al. (2013) and Acemoglu et al. (2016).

⁵² Davidson and Hadshar (2025) and Binder (2026) consider scenarios for rapid robot automation. Denain and Rivière (2026) focus on the first few years of robot production in a scenario where demand for humanoids and other robots far exceeds supply.

⁵³ Picking a functional form to extrapolate physical work shares requires heroic assumptions. From 1977 to 2026, the share of physical work fell about 0.27 percentage points per year. A log linear or logistic fit seems more appropriate than a linear one, which would turn negative in the future. The decline in physical share also slowed to 0.16 percentage points per year between 2008 and 2026 (0.16 using rounded shares in Appendix Figure 4), and a similar slowdown is present in the agriculture employment share (BLS 2026b). Nonetheless, even a linear trend predicts the share of physical work will reach zero only by the 2170s.